

EE/CE 252/251

Signals and Systems

with

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@

Habib University

Signal Basics II

Operations

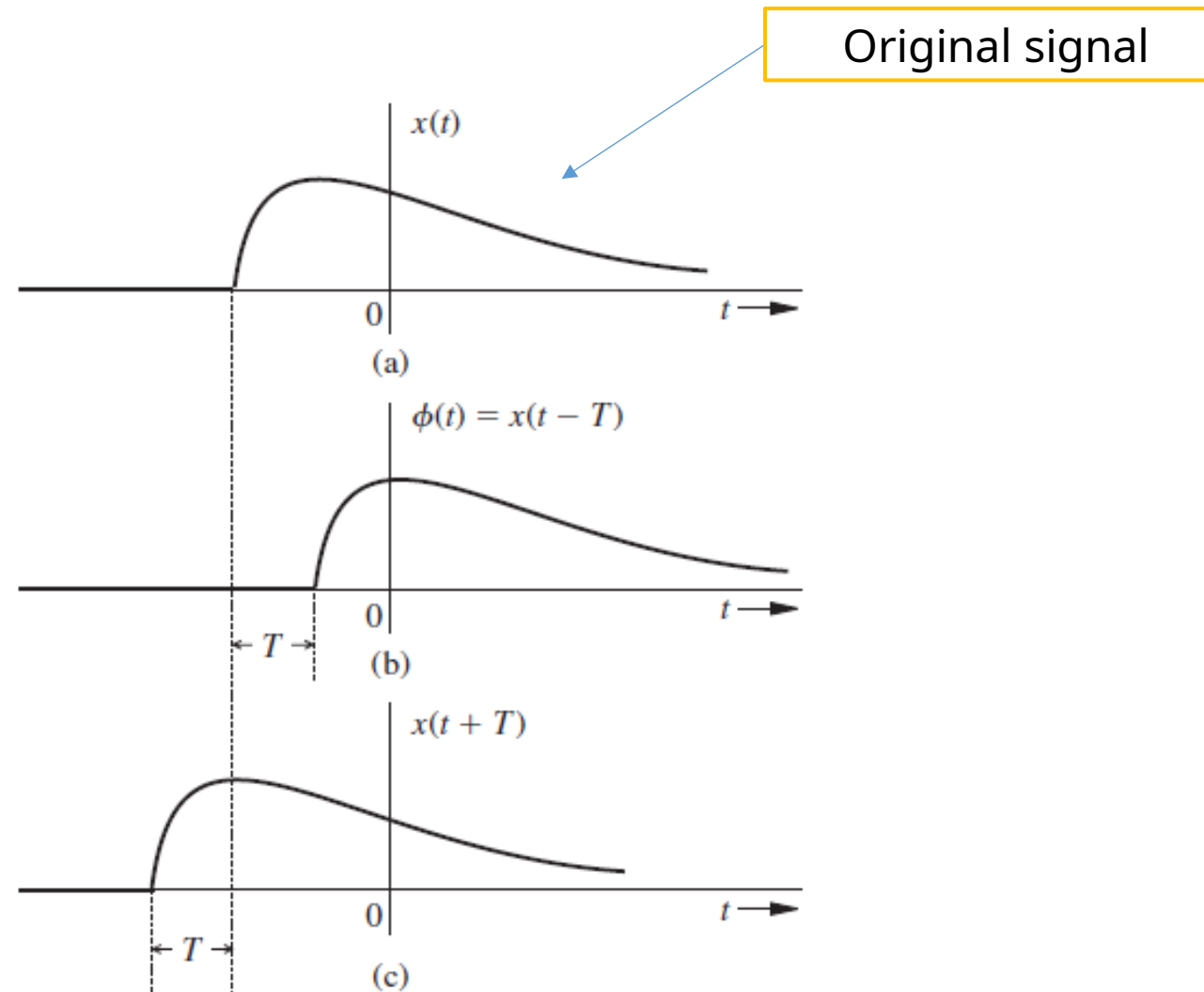
Messing with signals

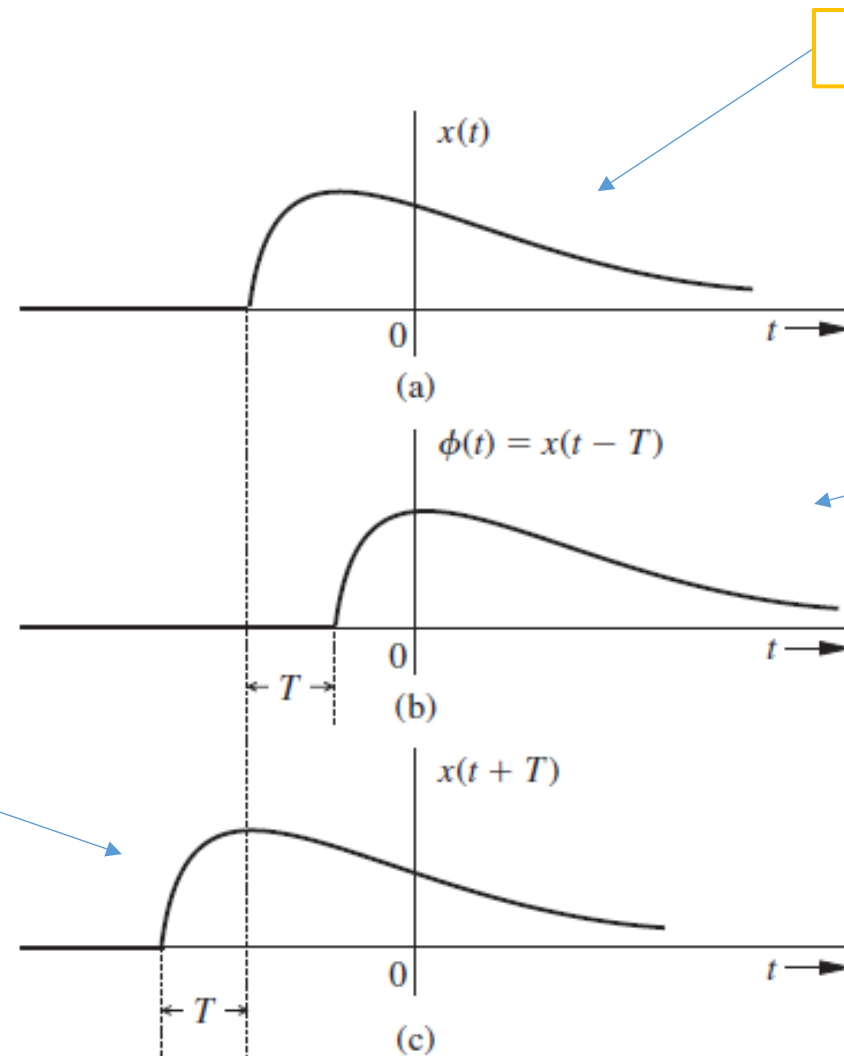
Examples

Some practice problems

Systems often alter signals. It is good to know some of these changes...

How are the two signals related to the original?



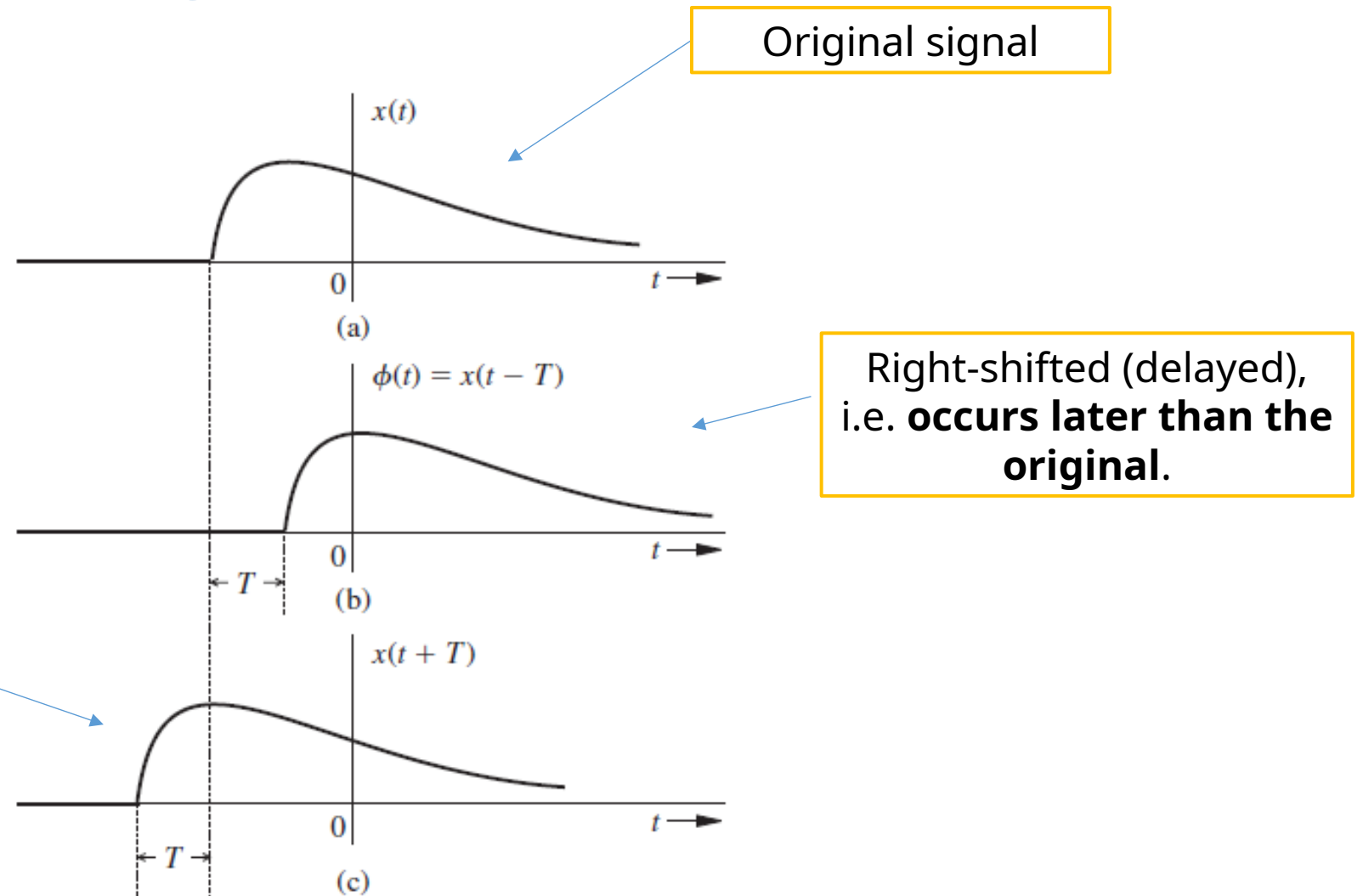


Original signal

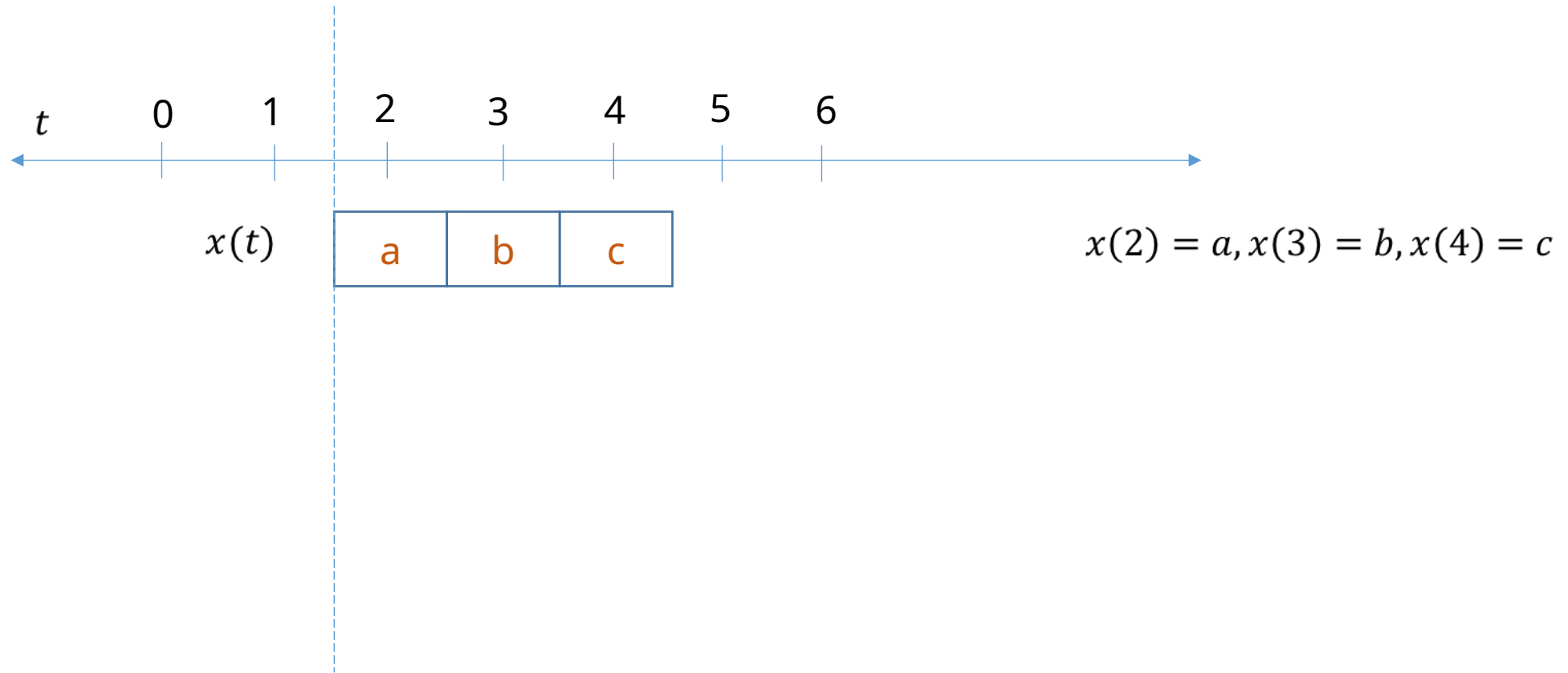
Right-shifted (delayed),
i.e. **occurs later than the original.**

Left-shifted (advanced),
i.e. **occurs earlier than the original.**

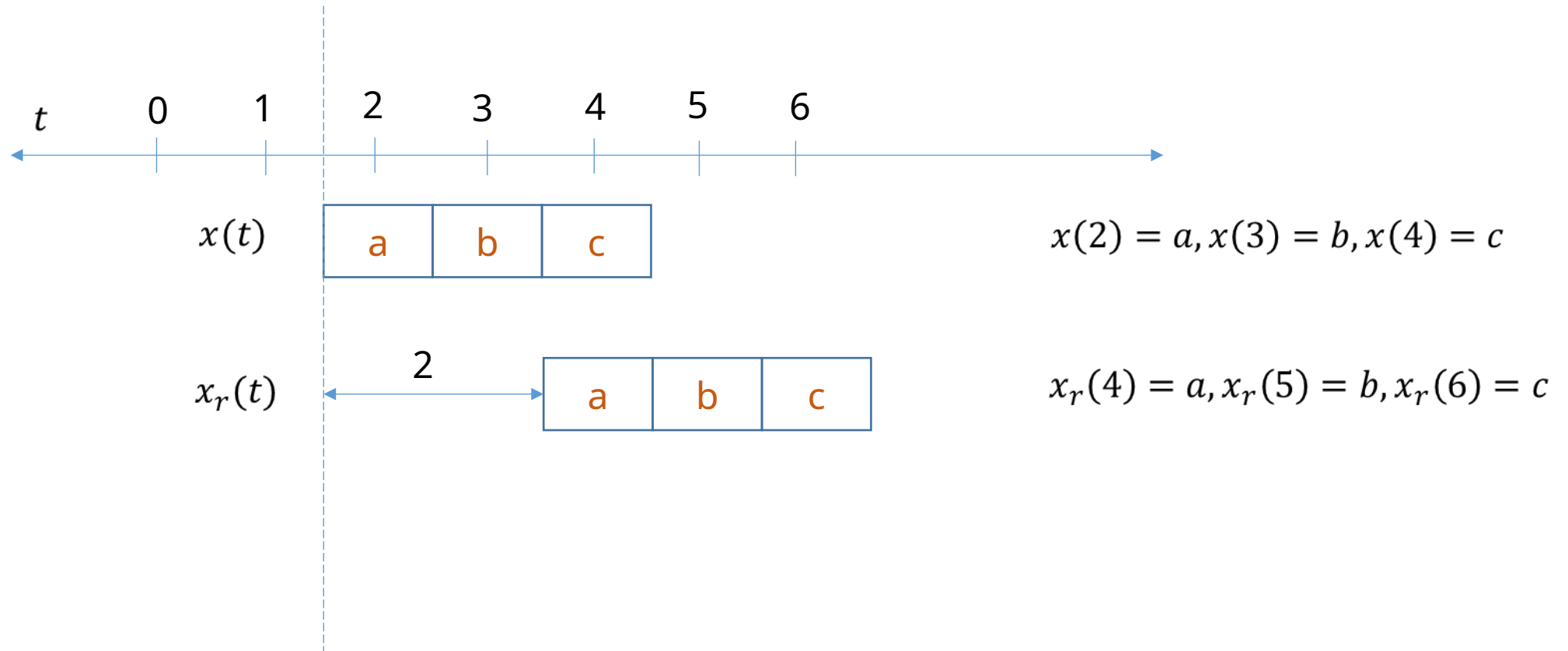
1. Time-Shifting



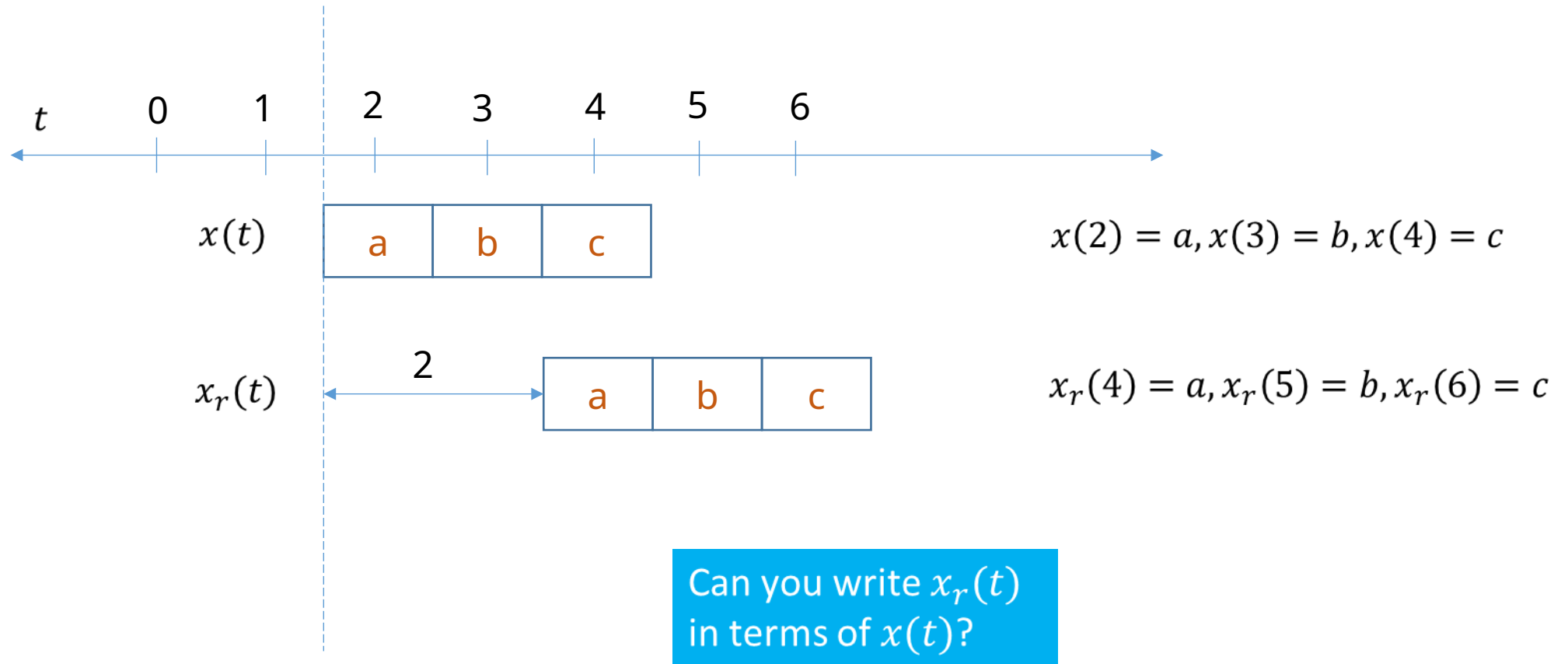
1. Time-Shifting: How to write mathematically?



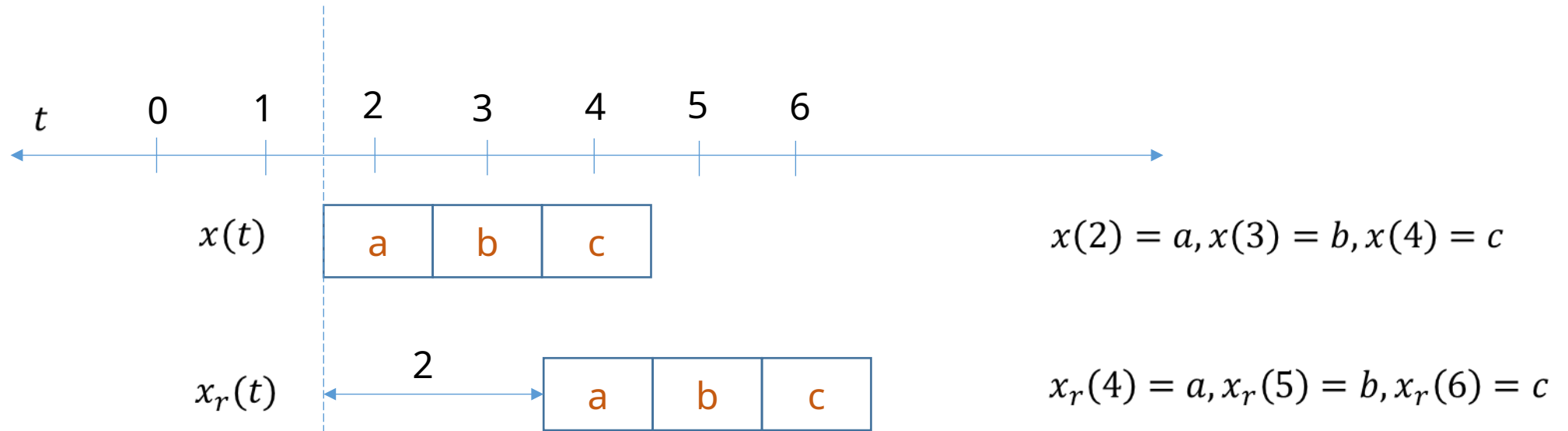
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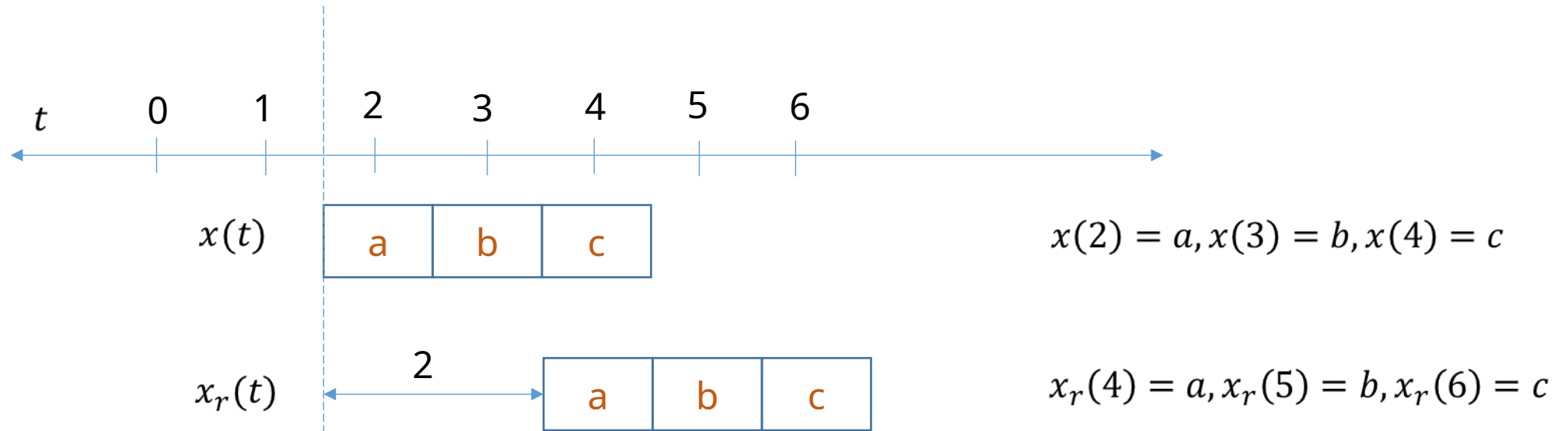
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Can you write $x_r(t)$
in terms of $x(t)$?

$$\begin{aligned}x_r(4) &= x(2) = a \\x_r(5) &= x(3) = b \\x_r(6) &= x(4) = c \\&\dots \\x_r(t) &= ?\end{aligned}$$

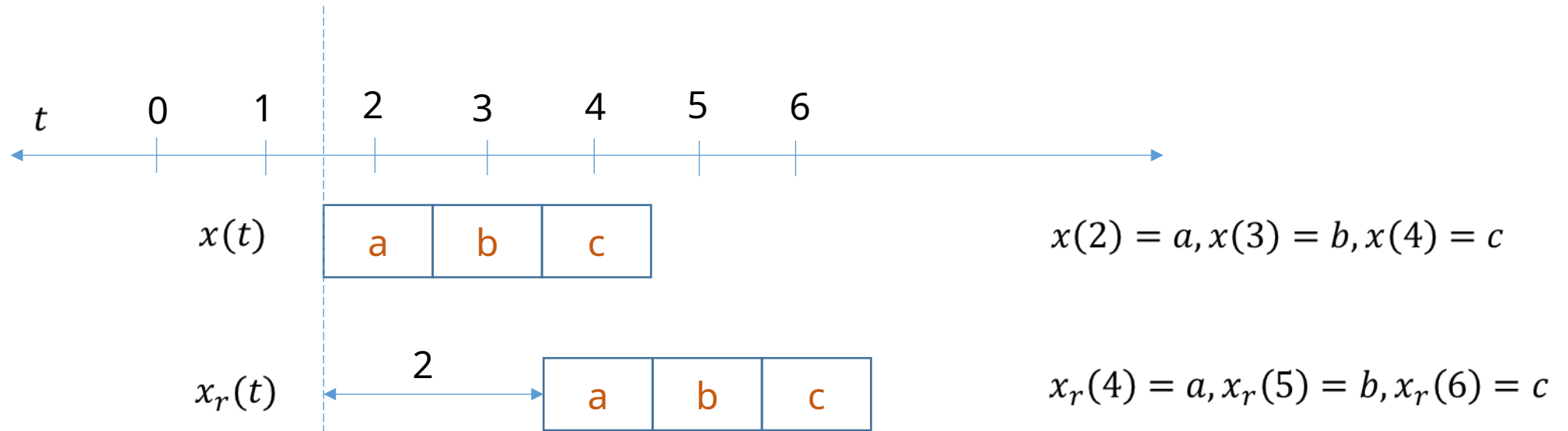
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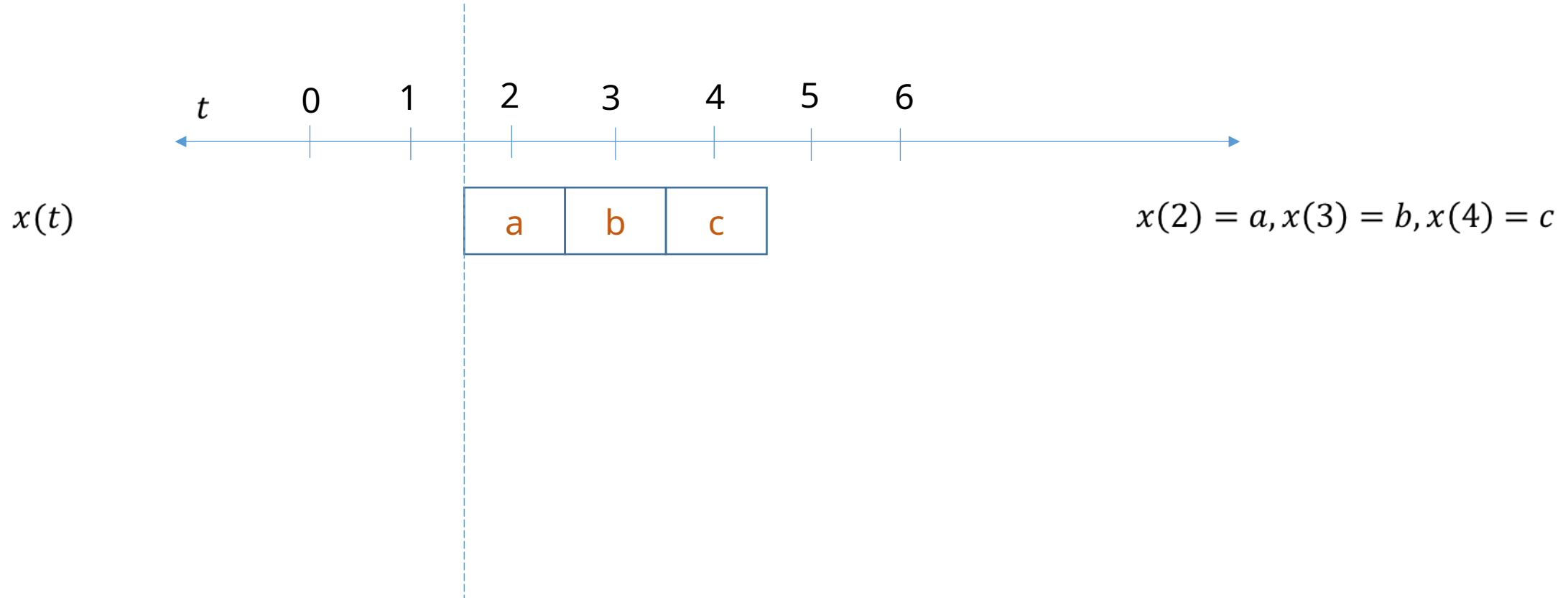


In general, for a right-shift of T units, we can write the new signal as $x(t - T), T \geq 0$

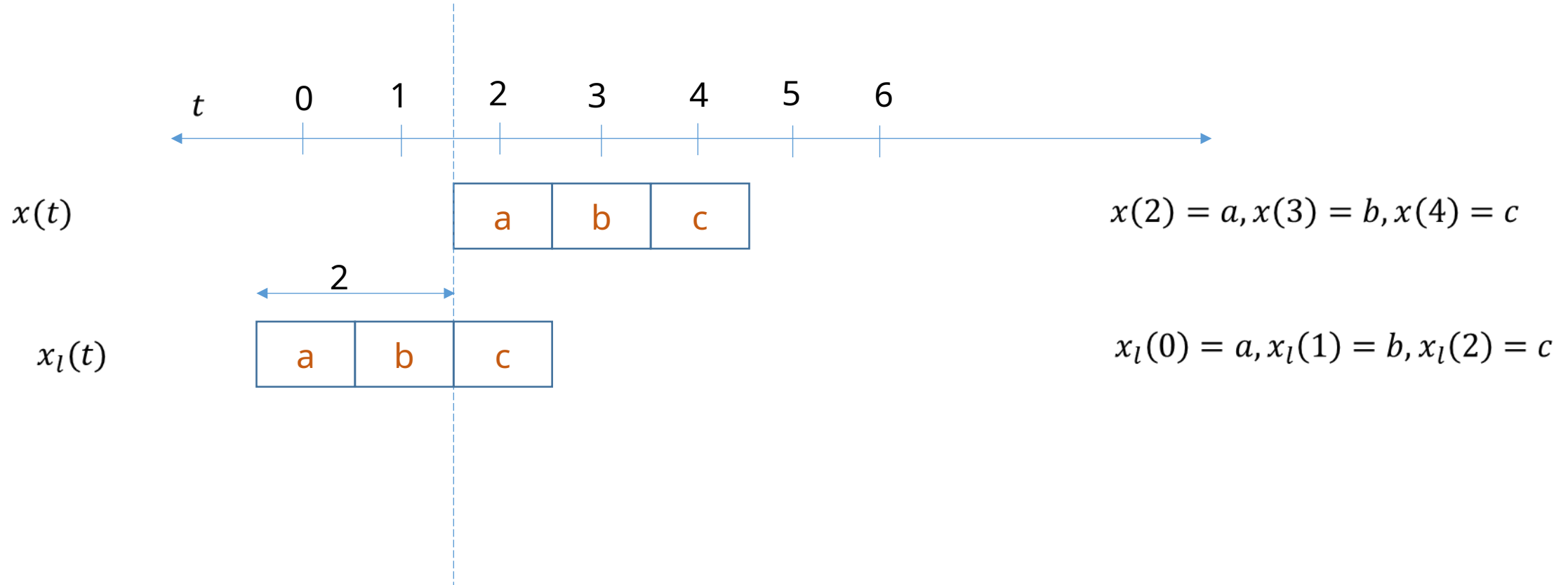
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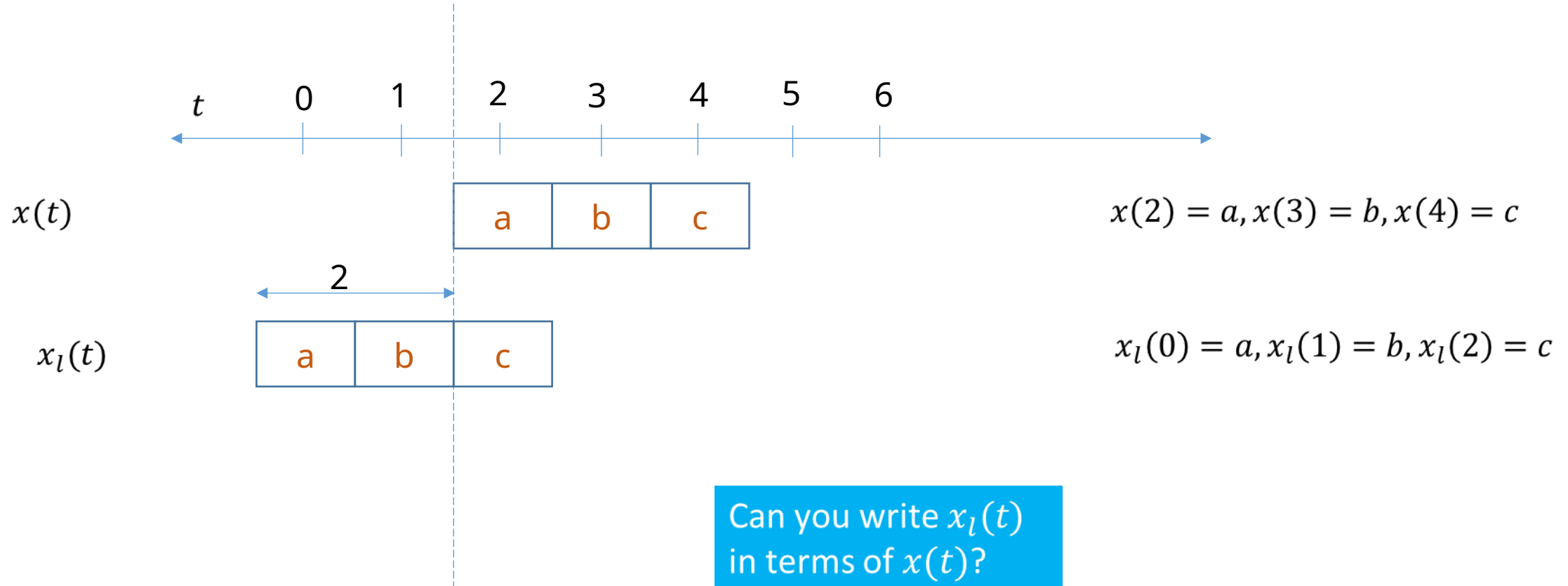
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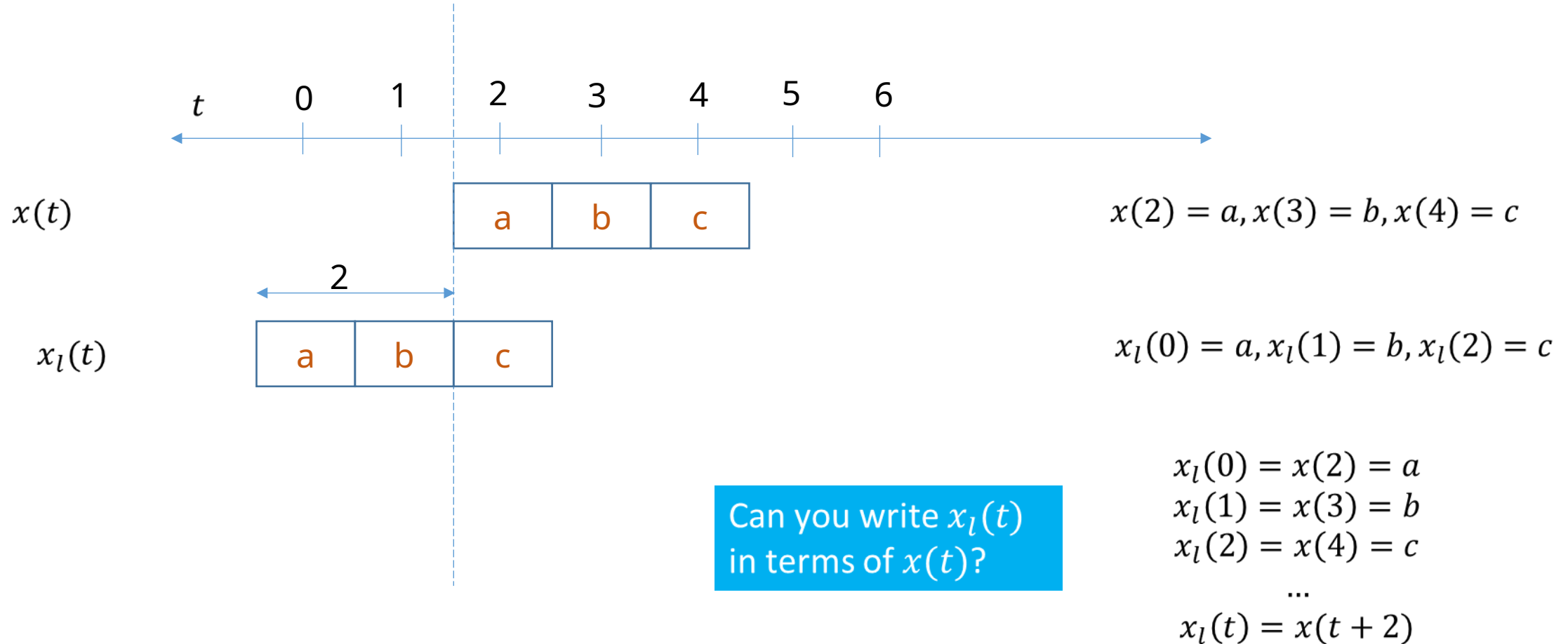
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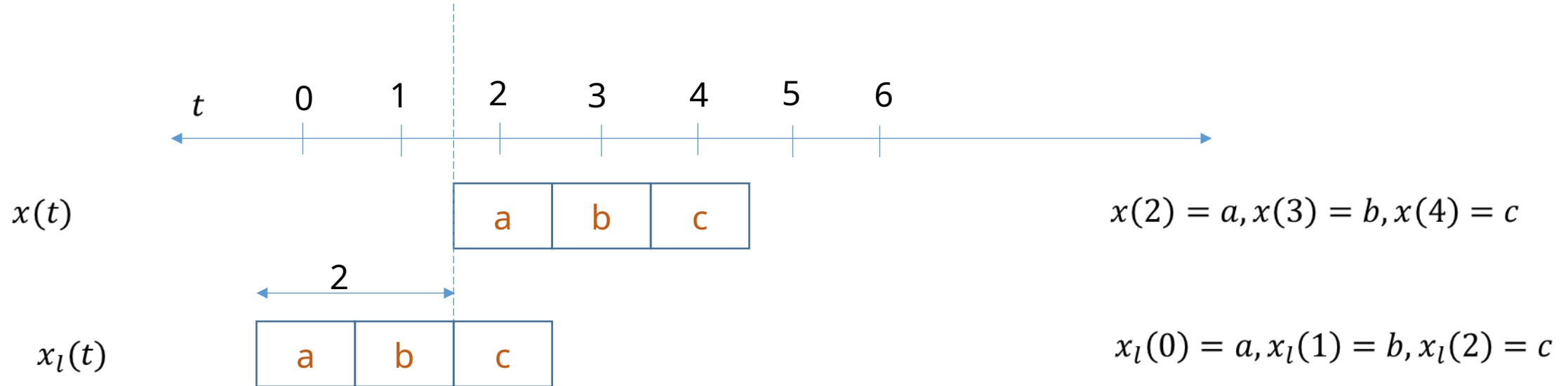
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In general, for a left-shift of T units, we can write the new signal as $x(t + T), T \leq 0$

Can you write $x_l(t)$ in terms of $x(t)$?

$$\begin{aligned} x_l(0) &= x(2) = a \\ x_l(1) &= x(3) = b \\ x_l(2) &= x(4) = c \\ &\dots \\ x_l(t) &= x(t + 2) \end{aligned}$$

Early or
Late?

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An easy way
to remember

The sign is counter-intuitive as in $x(t \pm T)$, $-$ is
for delay and $+$ is for advance (as long as $T > 0$)

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The sign is counter-intuitive as in $x(t \pm T)$, $-$ is for delay and $+$ is for advance (as long as $T > 0$)

Quick numeric check

Put $t = 0$ in $x_{new}(t) = x(t - 1)$ to see that whatever happened at $x(-1)$ is now happening at $x_{new}(0)$.

1. Time-Shifting: Example

$$x(t) = \begin{cases} e^{-2t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

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Step 2 $x(t - 1) = \begin{cases} e^{-2(t-1)} \\ 0 \end{cases}$ Anything missing?

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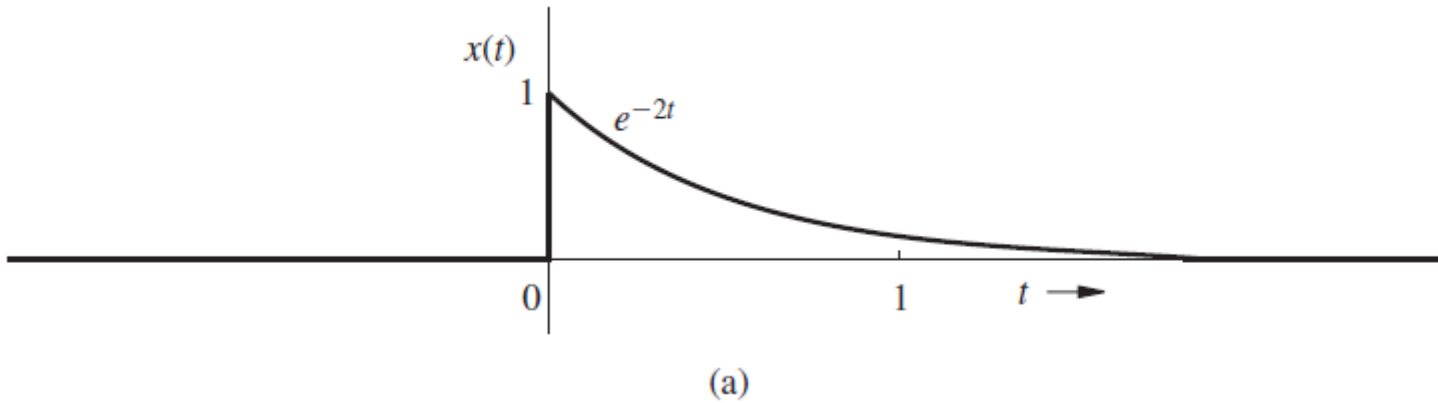
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Can you plot these signals?

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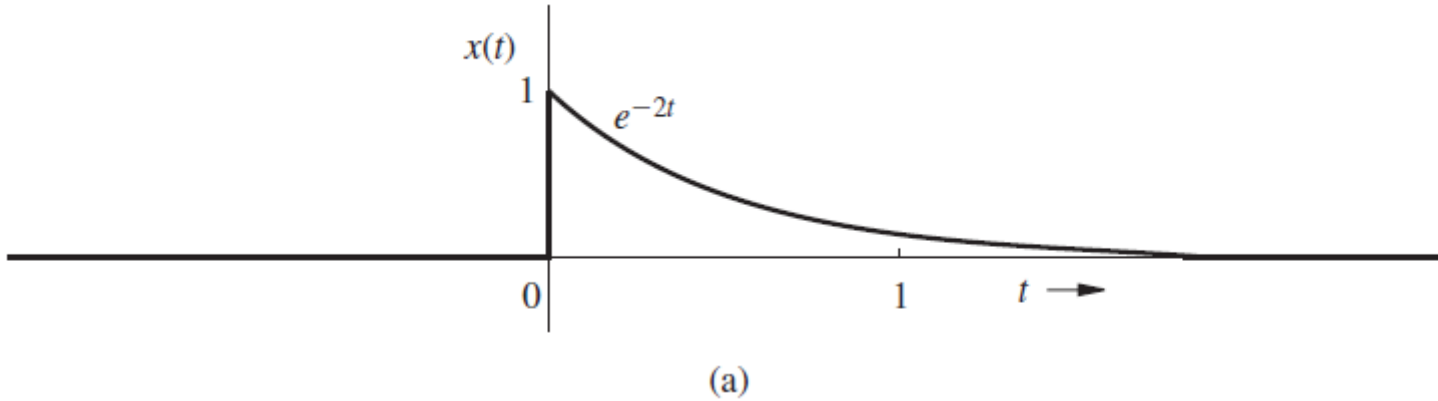
1. Time-Shifting: Example (plotting)



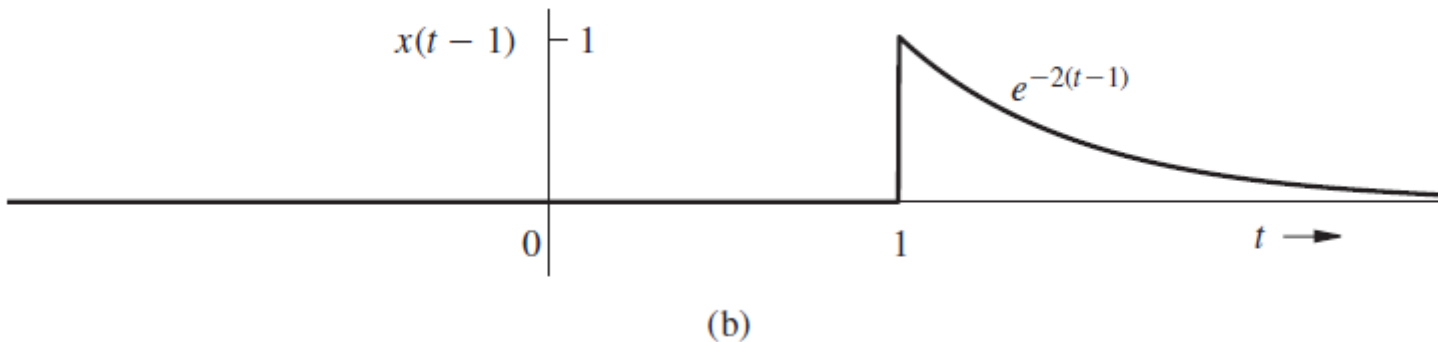
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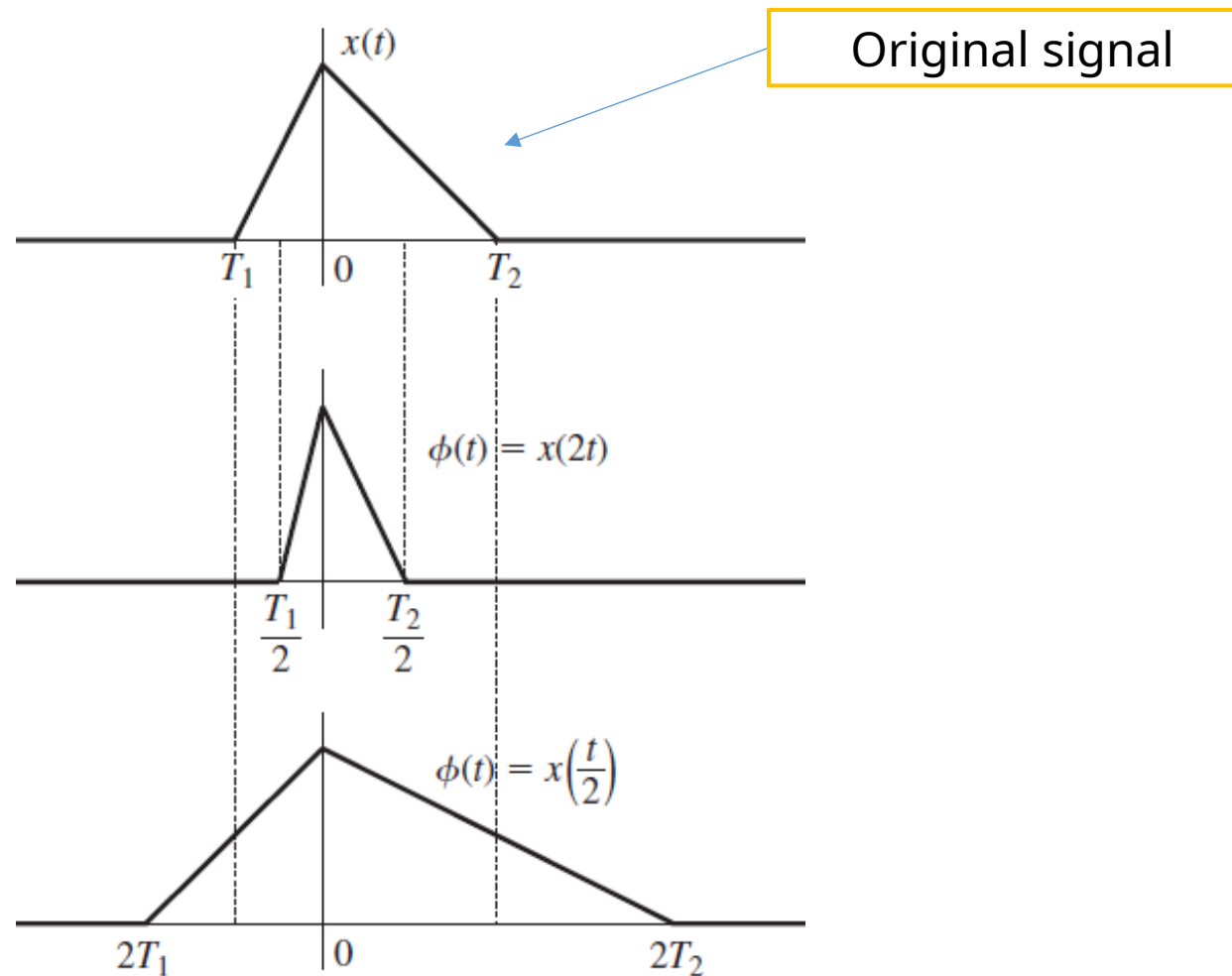


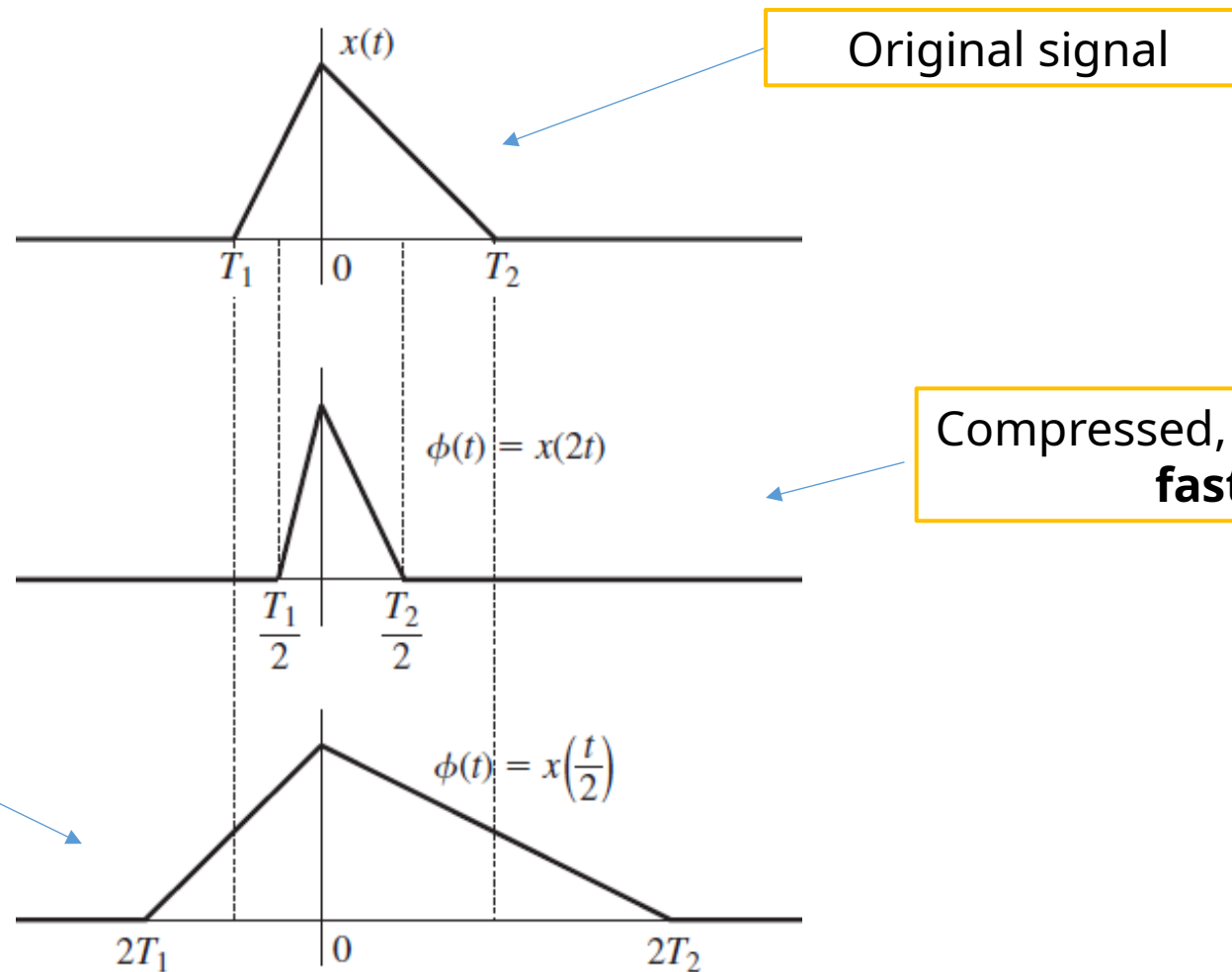
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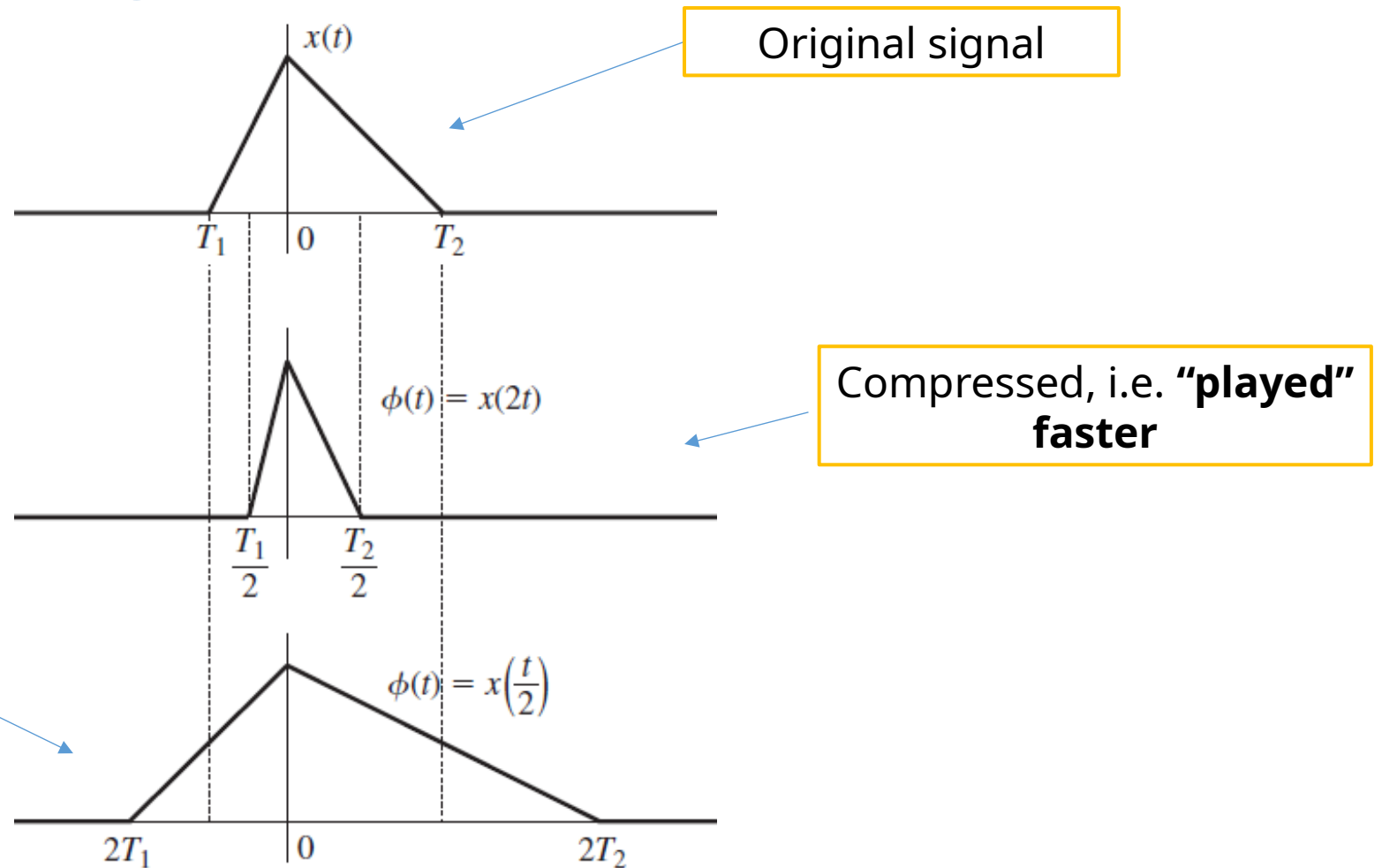


Original signal

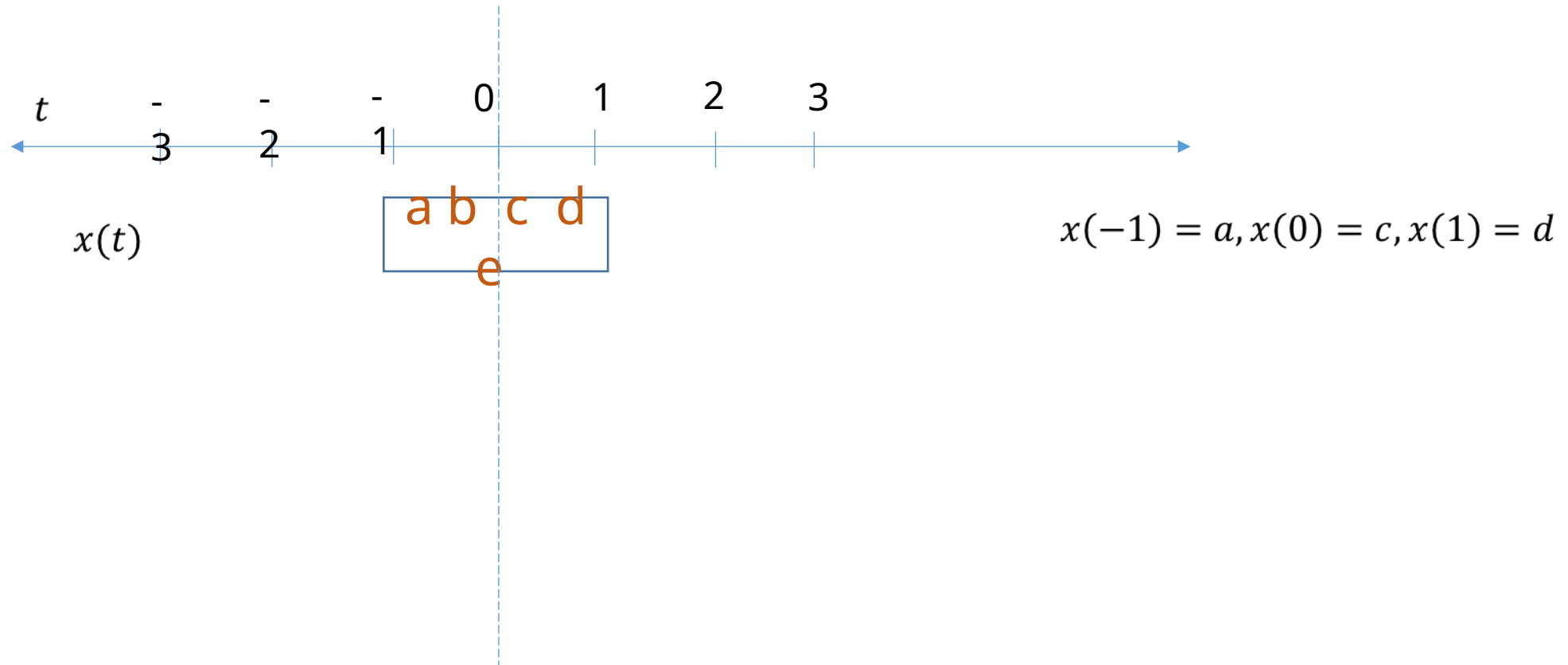
Compressed, i.e. **“played”
faster**

Expanded, i.e. **“played”
slower**

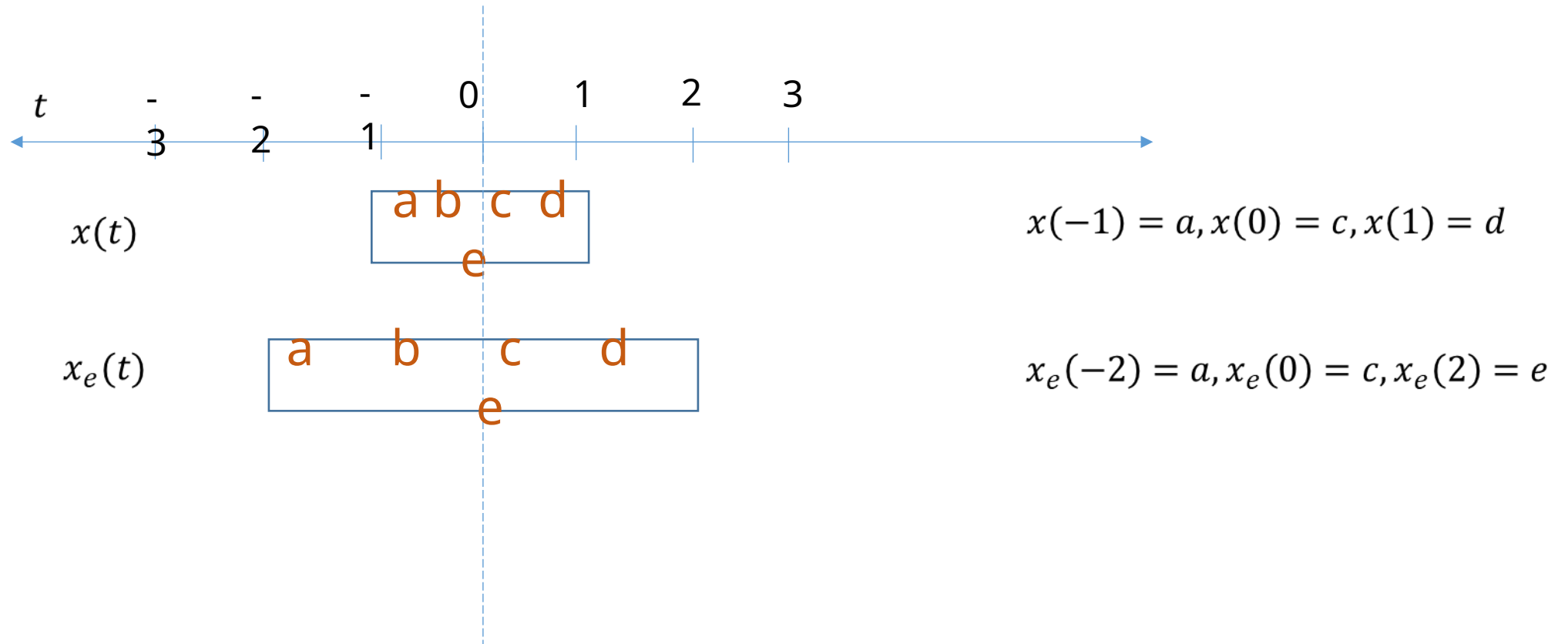
2. Time-Scaling



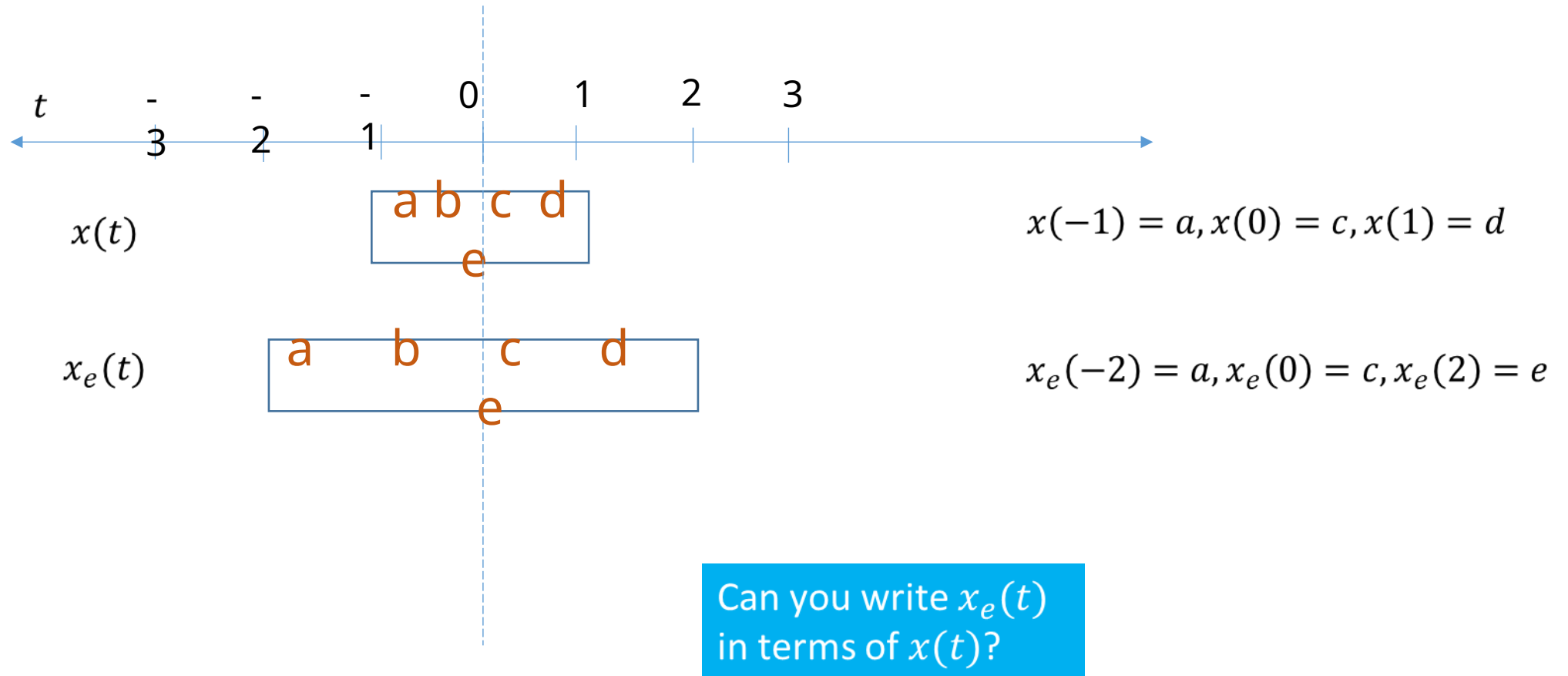
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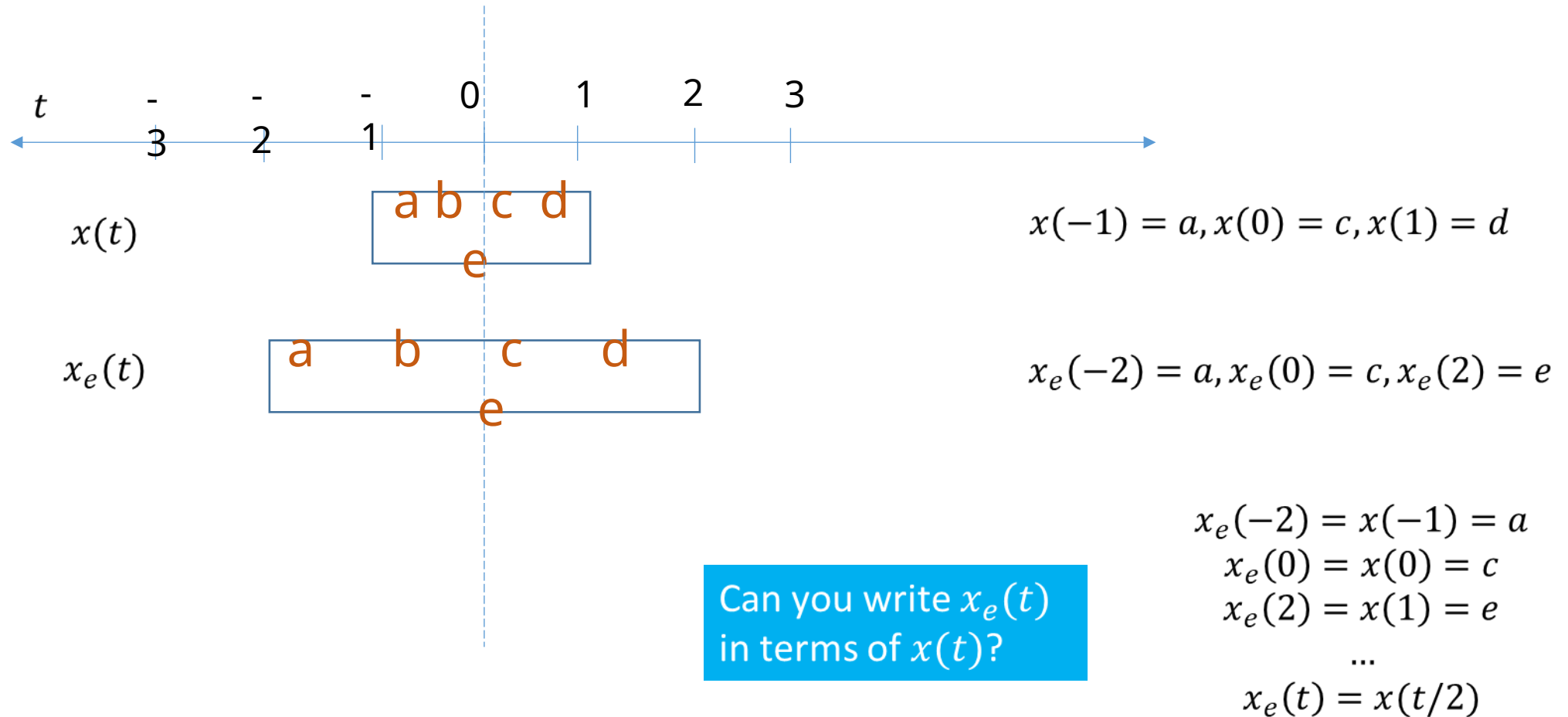
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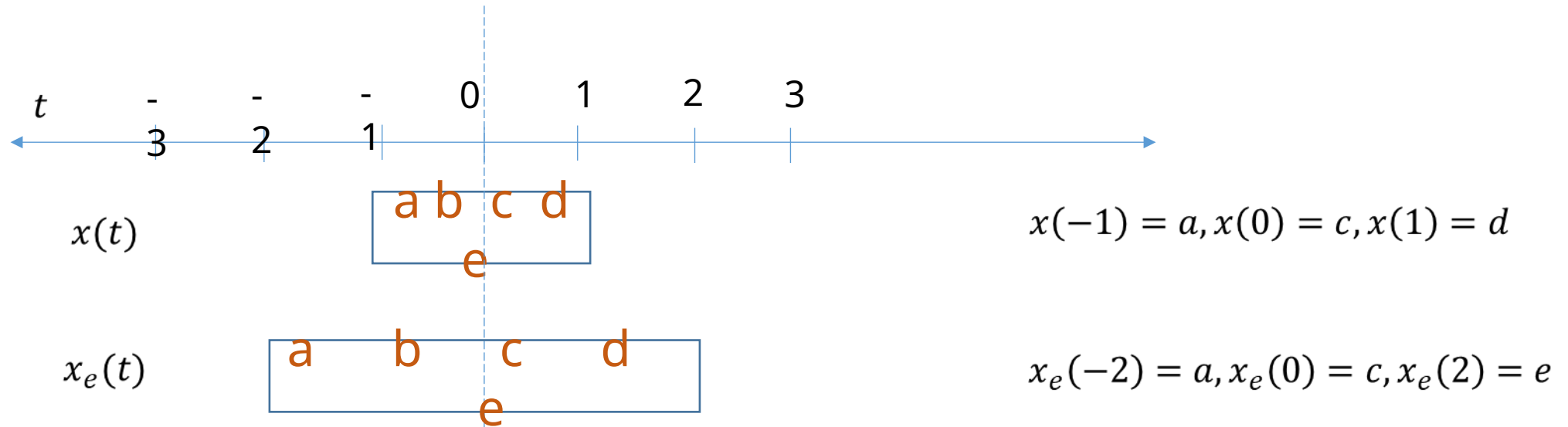
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In general, a time-expanded version of a signal can be written as

$$x\left(\frac{t}{a}\right), a > 1$$

Can you write $x_e(t)$ in terms of $x(t)$?

$$\begin{aligned}
 x_e(-2) &= x(-1) = a \\
 x_e(0) &= x(0) = c \\
 x_e(2) &= x(1) = e \\
 &\dots \\
 x_e(t) &= x(t/2)
 \end{aligned}$$

2. Time-Scaling: How to write mathematically?

In general, for a time-scaling (**expansion**) by a factor $a > 1$, we can write the expanded signal as

$$x(t/a)$$

And for a time-scaling (**compression**) by a factor $a > 1$, we can write the compressed signal as

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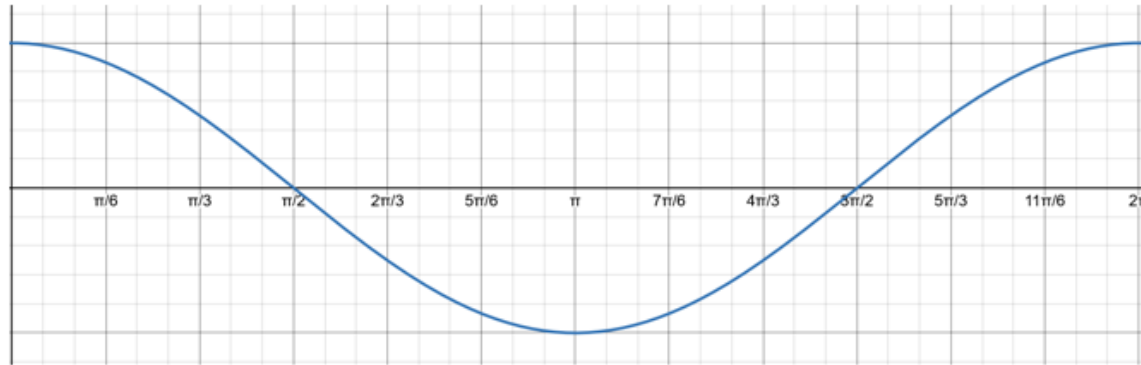
$$x(at)$$

Interestingly, this also shows why $\cos(2t)$ has a higher frequency than $\cos(t)$ – something we saw earlier...

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Visualizing relation between t and θ

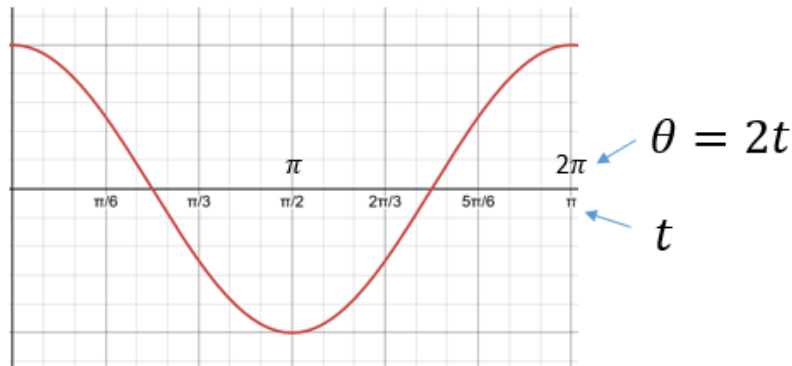
$\cos(t)$ has period $T_0 = 2\pi$ and frequency $f_0 = 1/2\pi$



$\theta = t$

As time goes from 0 to 2π , the angle θ also goes from 0 to 2π , and $\cos(t)$ completes one cycle.

Recall



$\cos(2t)$ has period $T_0 = \pi$ and frequency $f_0 = 1/\pi$

As time goes from 0 to π , the angle θ goes from 0 to 2π , thus $\cos(2t)$ completes one cycle much faster.

Compressed or
Expanded?

Compressed or Expanded?

An easy way to remember

The operation seems is counter-intuitive as in $x(at)$ and $x\left(\frac{t}{a}\right)$, $\times$ is for compression and $\div$ is for expansion (as long as $a > 1$)

Compressed or Expanded?

An easy way to remember

The operation seems counter-intuitive as in $x(at)$ and $x\left(\frac{t}{a}\right)$, $\times$ is for compression and $\div$ is for expansion (as long as $a > 1$)

Quick numeric check

Put $t = 0$ and 4 in $x_{new}(t) = x(2t)$ to see that whatever happened between $x(0)$ and $x(8)$ is now happening between $x_{new}(0)$ and $x_{new}(4)$.

2. Time-Scaling: Example

$$x(t) = \begin{cases} 2 & -1.5 \leq t < 0 \\ 2e^{-t/2} & 0 \leq t < 3 \\ 0 & \text{otherwise} \end{cases}$$

Q. Write a compressed version of the signal, with compression factor $a = 3$.

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Anything missing?

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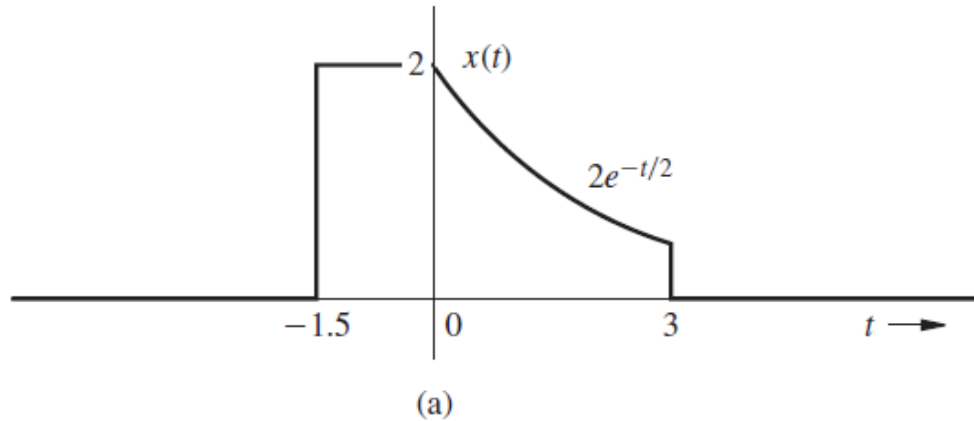
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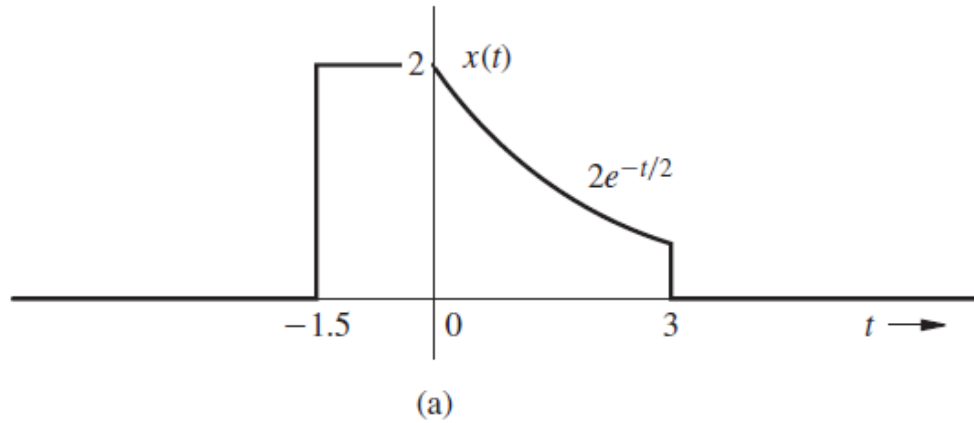
2. Time-Scaling: Example (plotting)



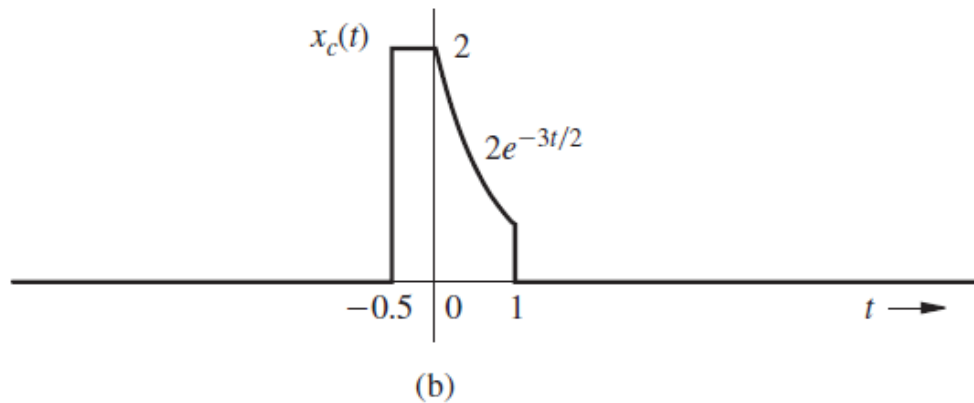
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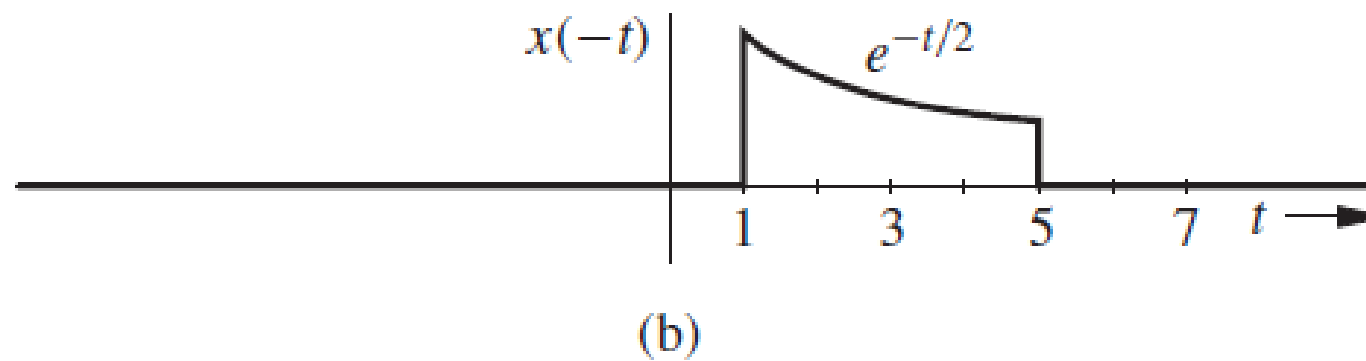
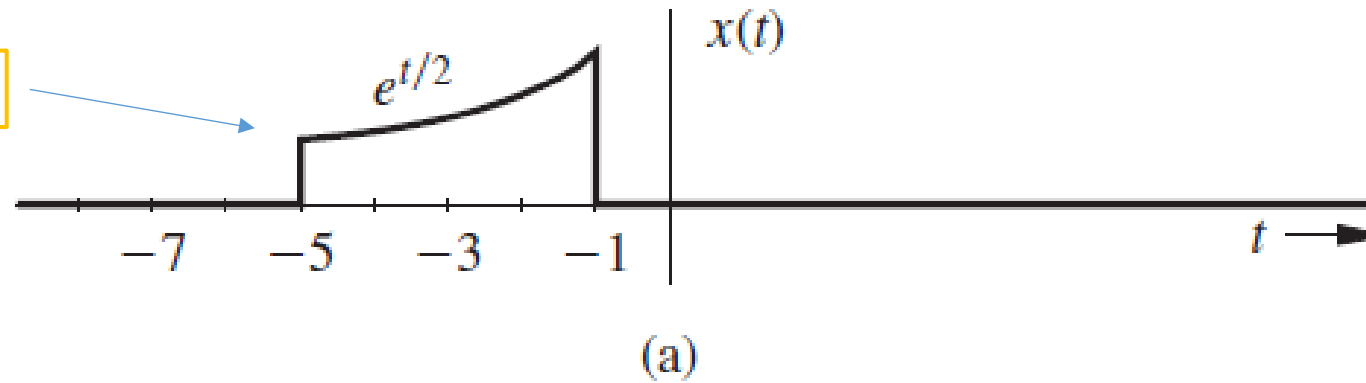
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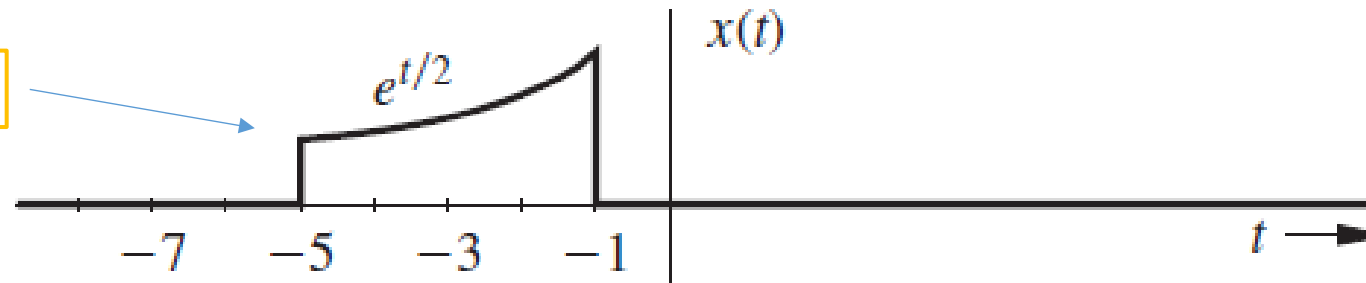
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related?

Original signal

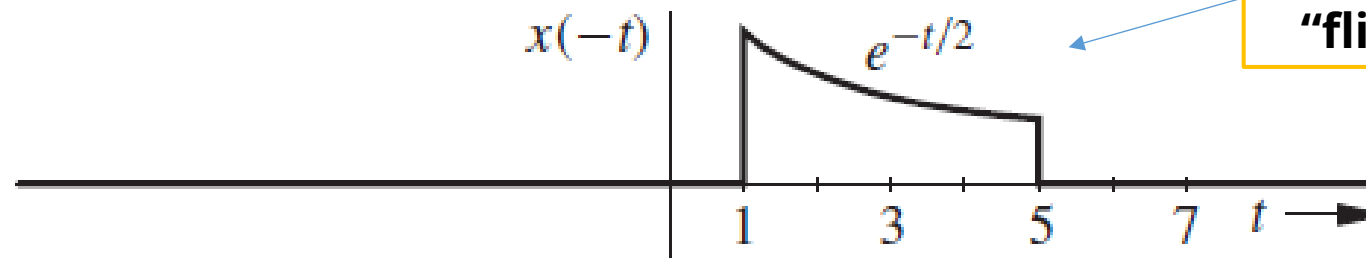


Original signal



(a)

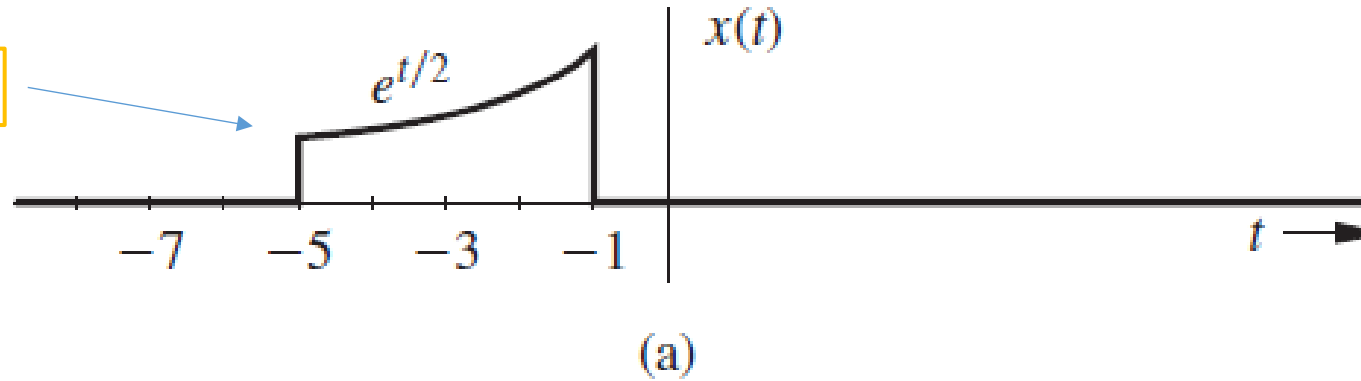
Time-reversed, i.e.
"flipped" horizontally



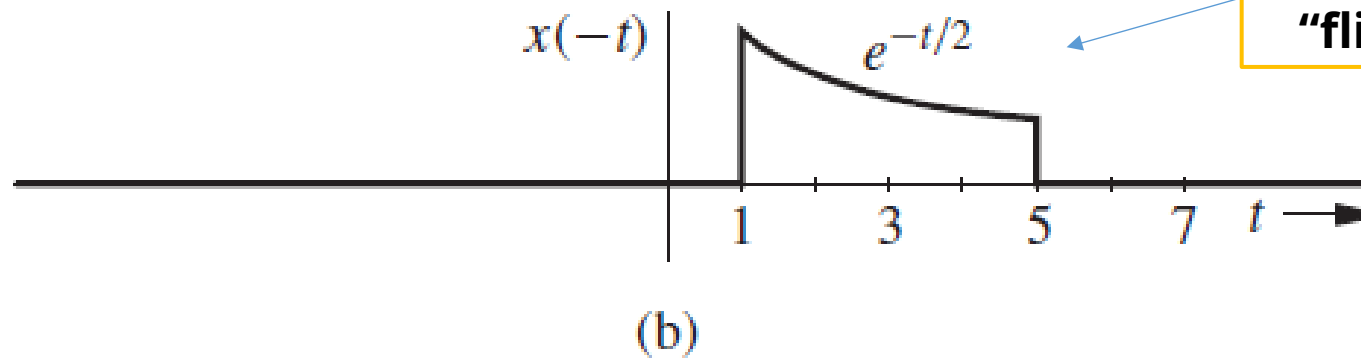
(b)

3. Time-Reversal

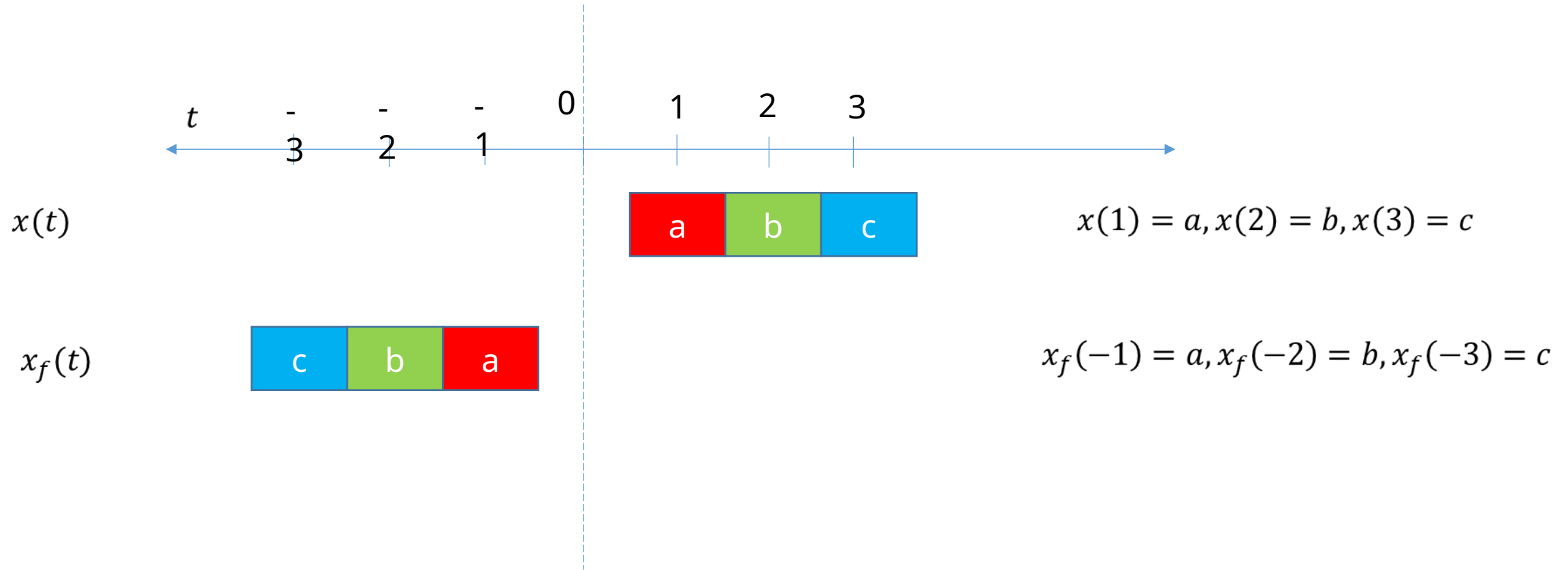
Original signal



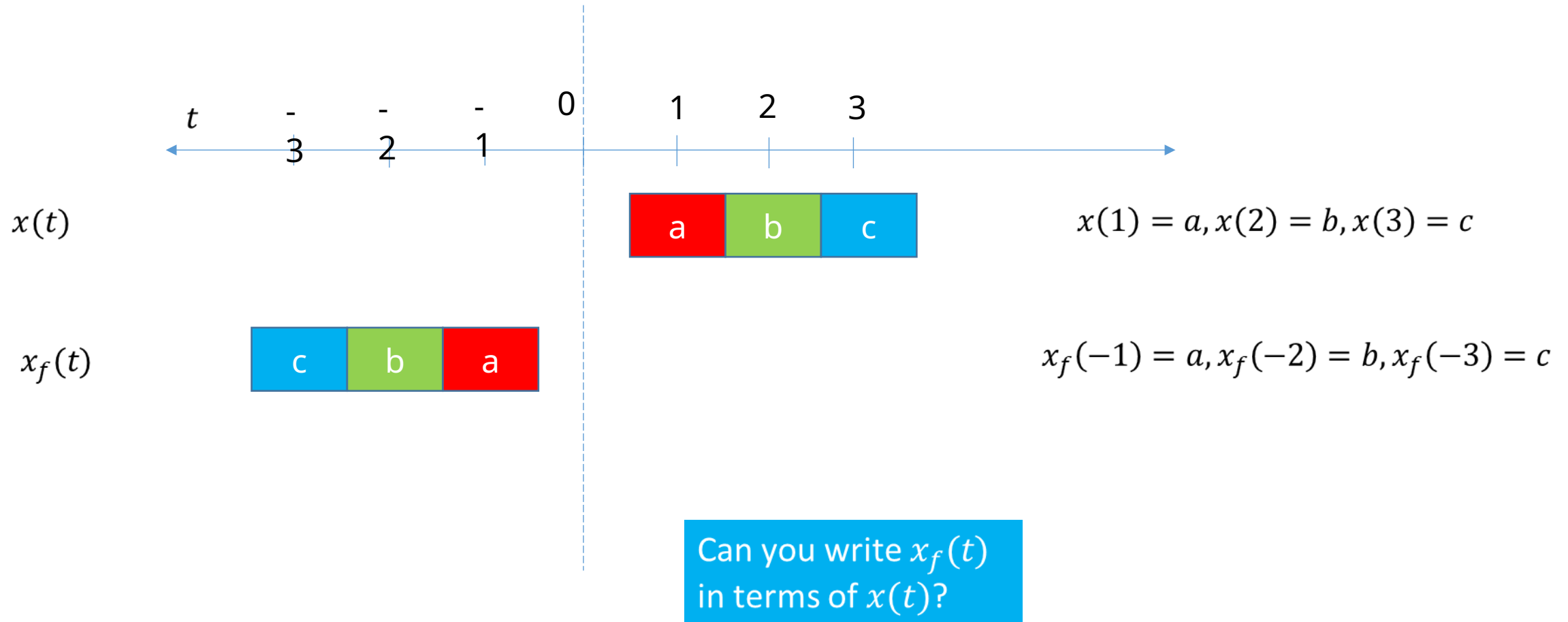
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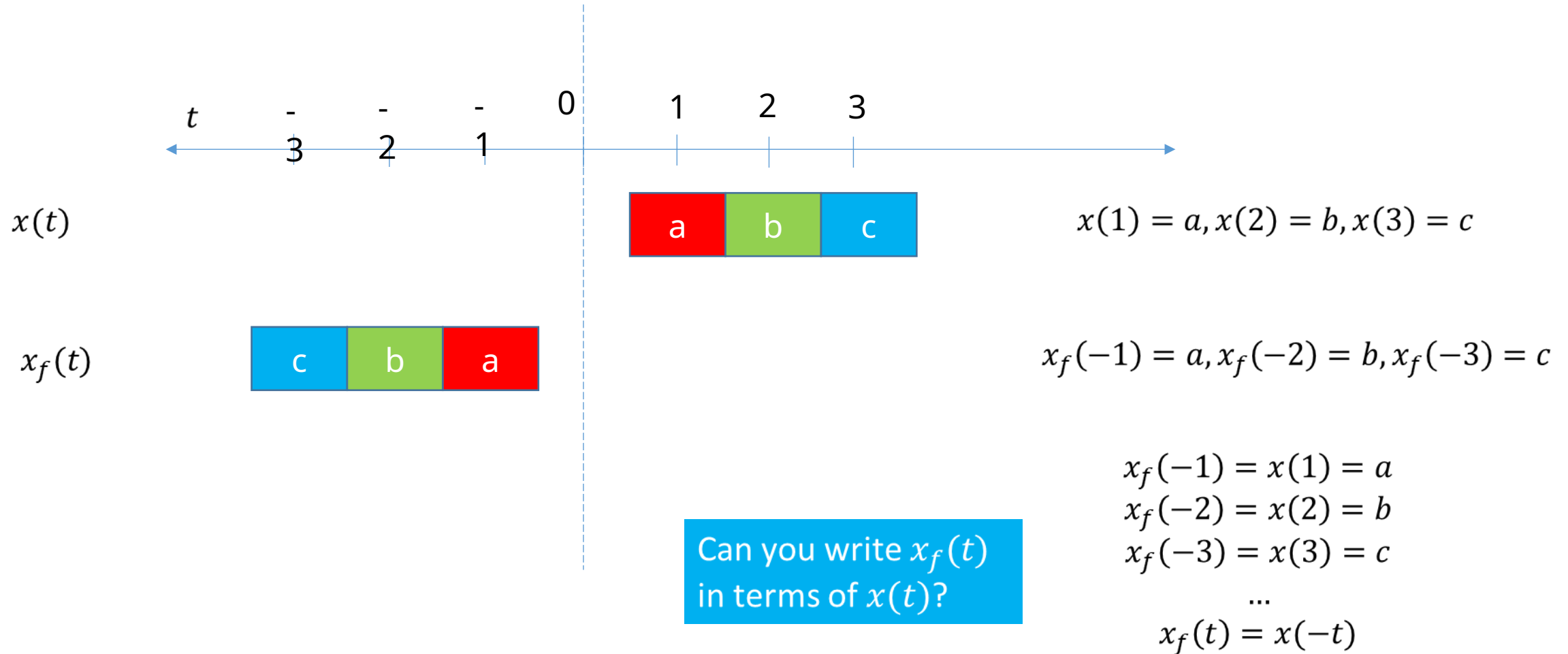
3. Time-Reversal: How to write mathematically?



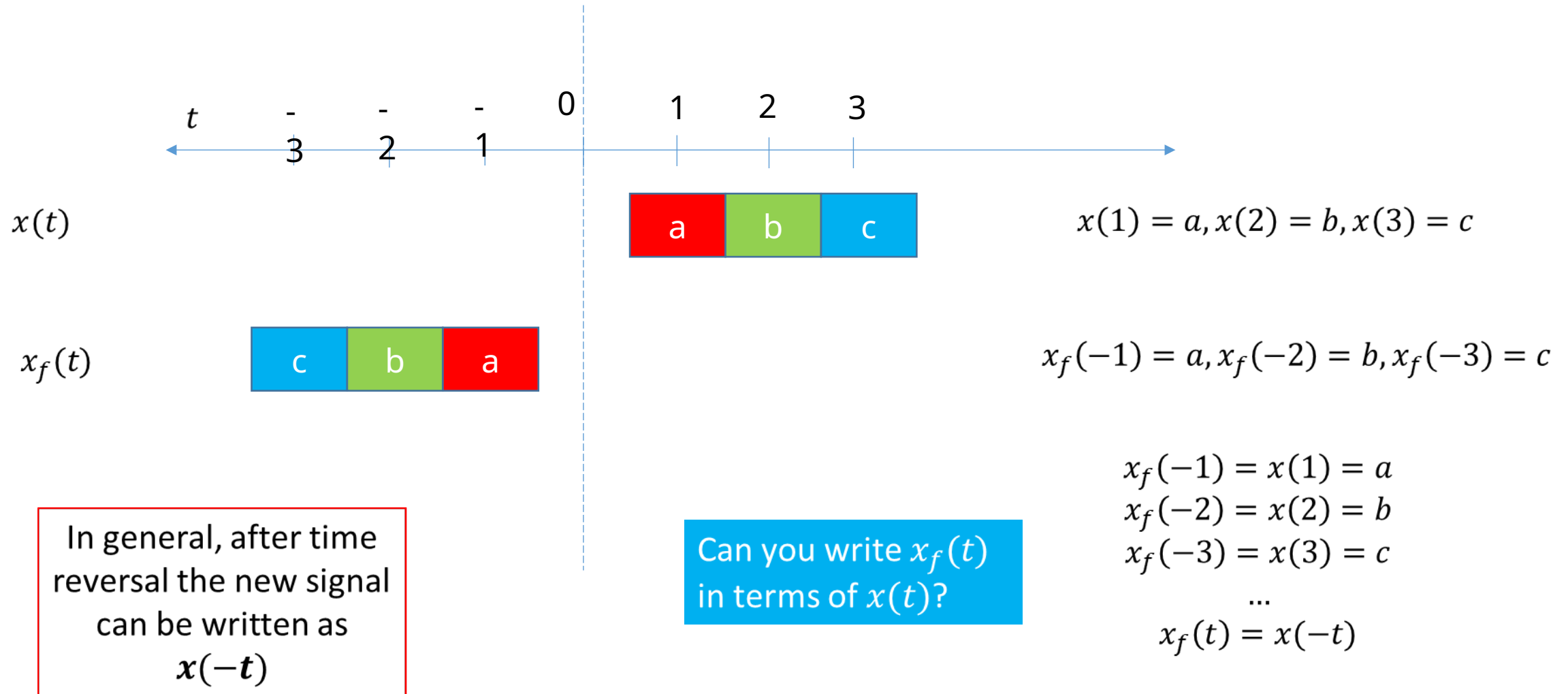
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3. Time-Reversal: Example

$$x(t) = \begin{cases} e^{t/2} & -1 \geq t > -5 \\ 0 & \text{otherwise} \end{cases}$$

Q. Write the time-reversed version of the given signal.

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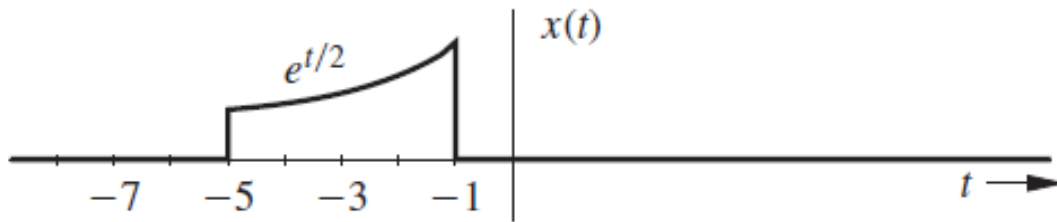
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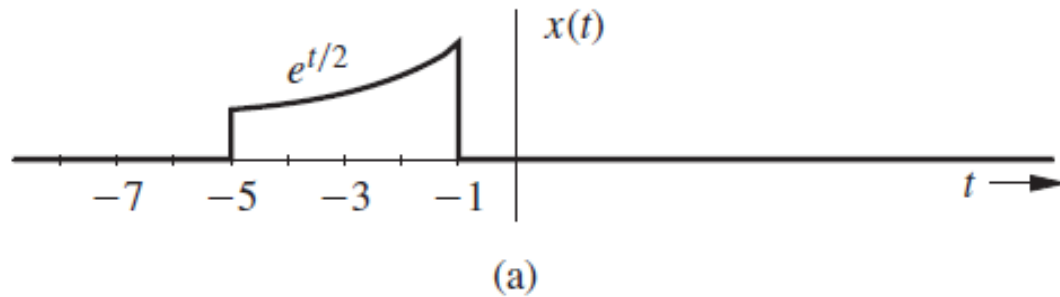
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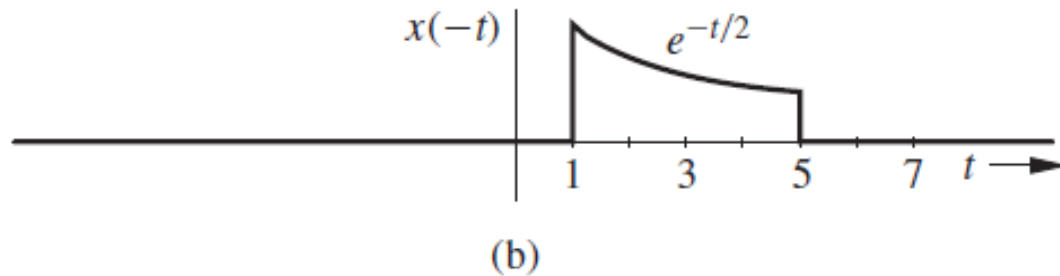
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3. Time-Reversal: Example (plotting)

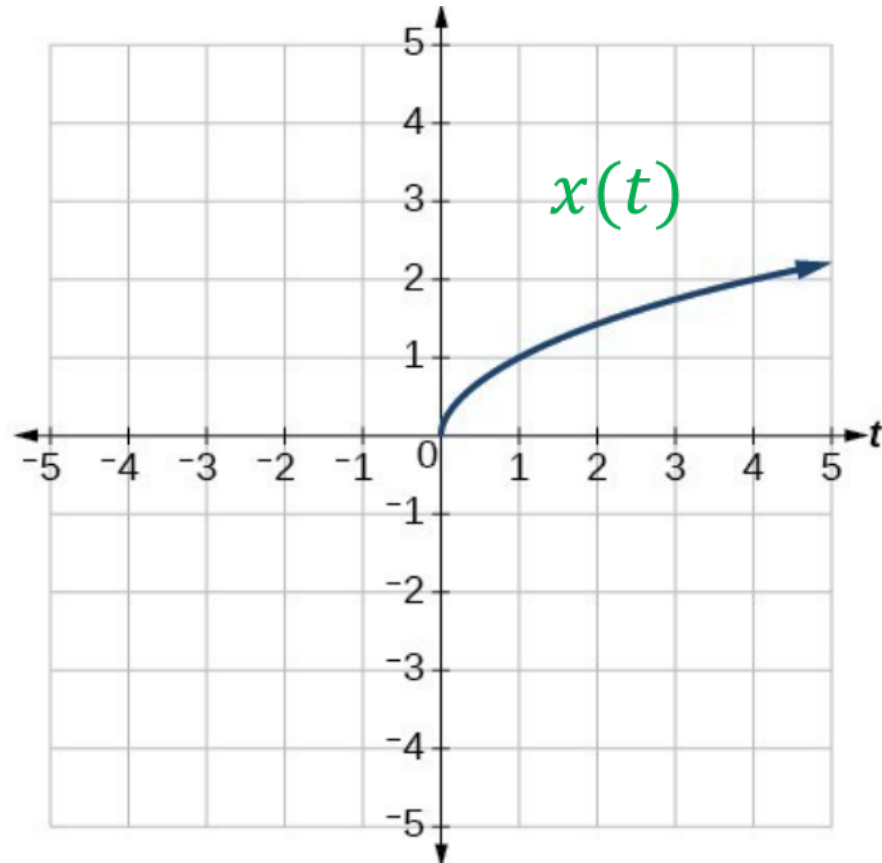


$$x(t) = \begin{cases} e^{t/2} & -1 \geq t > -5 \\ 0 & \text{otherwise} \end{cases}$$



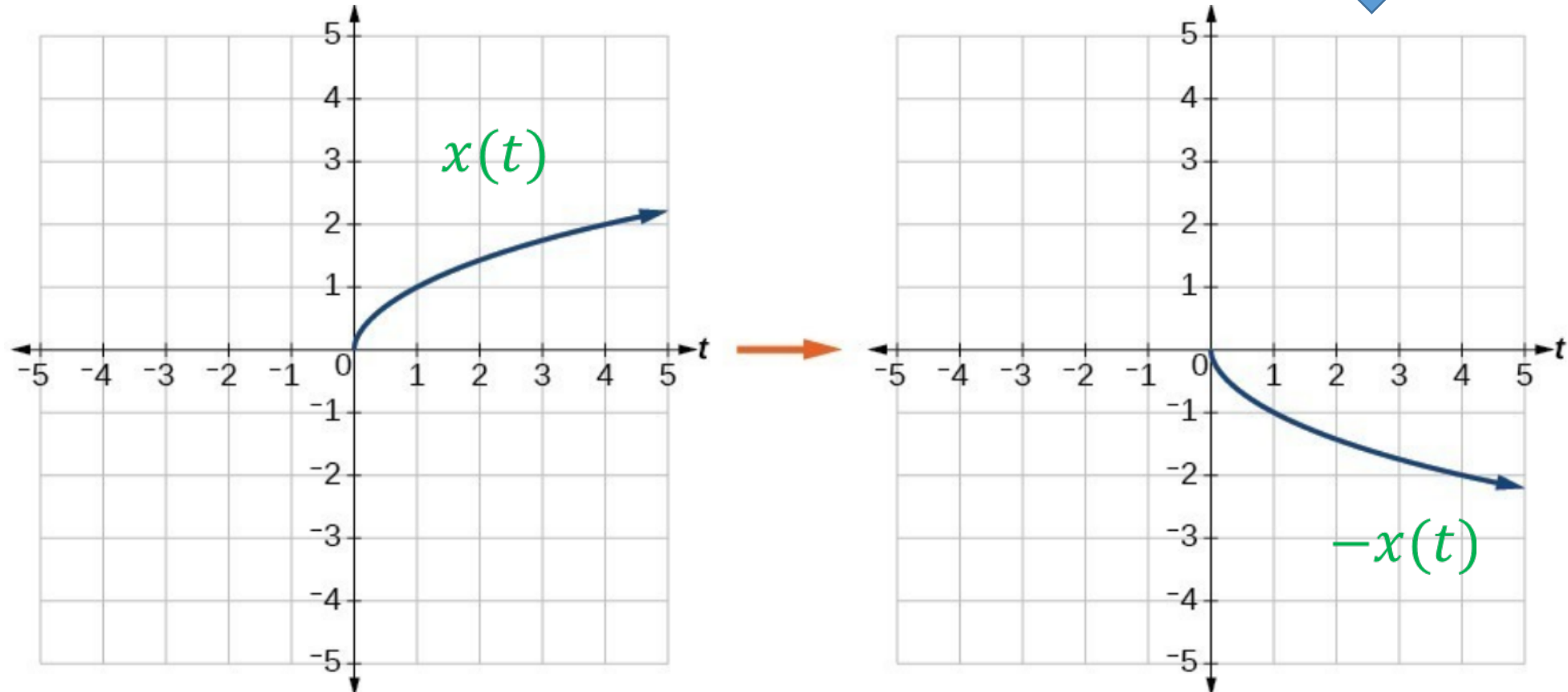
$$x(-t) = \begin{cases} e^{-t/2} & -1 \geq -t > -5 \quad \text{or} \quad 1 \leq t < 5 \\ 0 & \text{otherwise} \end{cases}$$

Q. What if we wanted to flip along the x -axis?



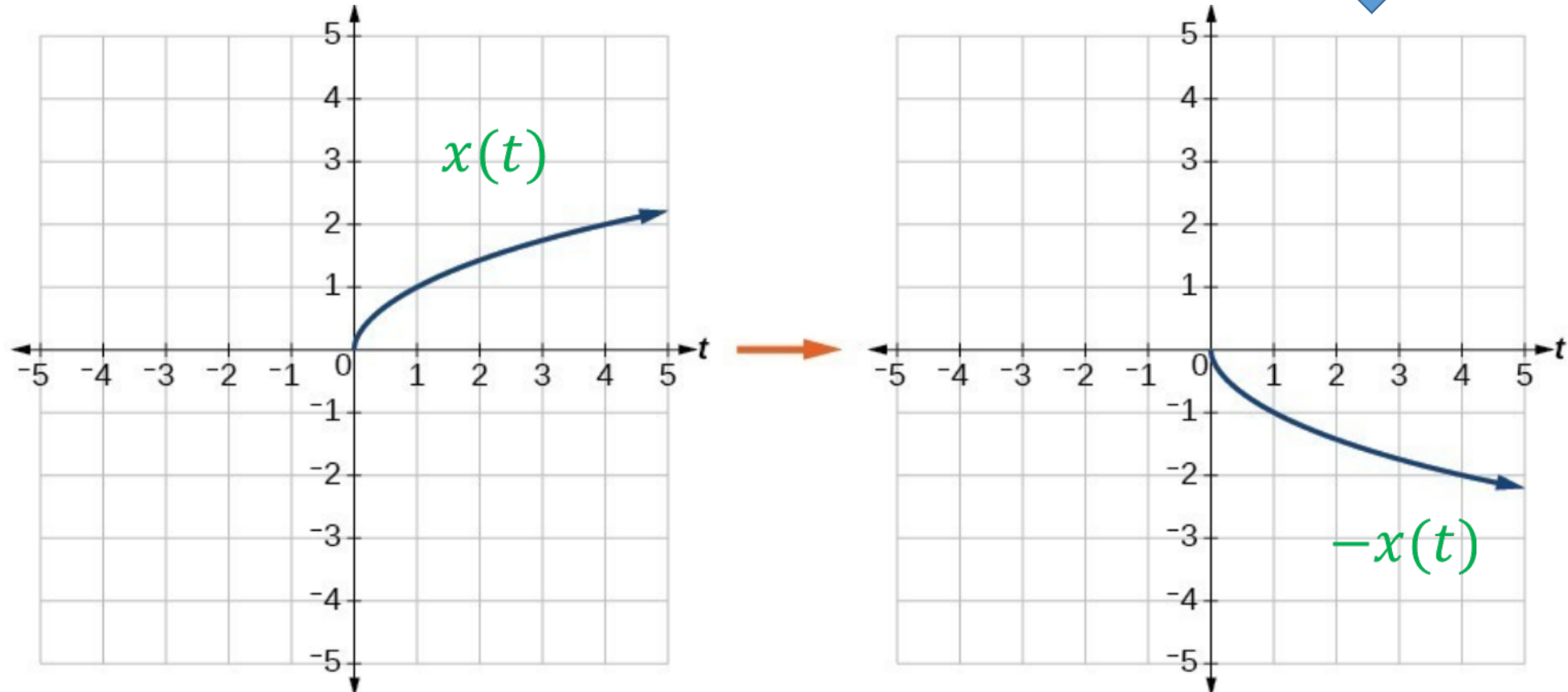
Q. What if we wanted to flip along the x -axis?

Amplitude-Reversal



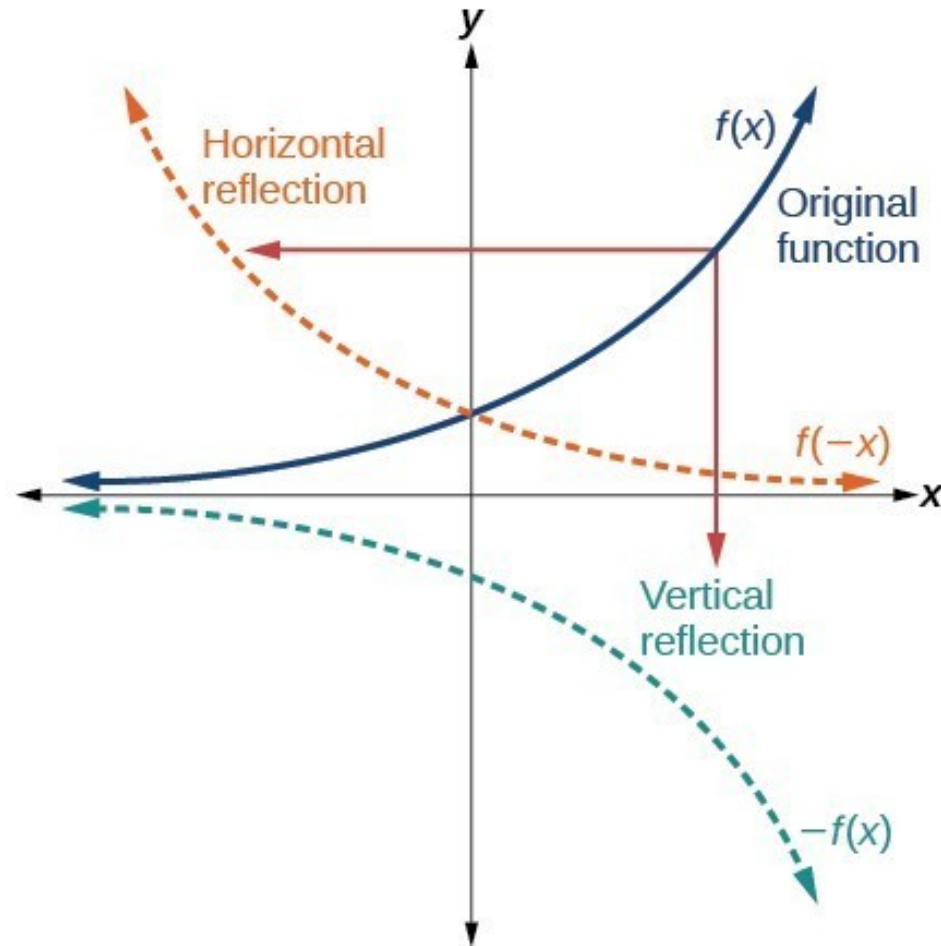
Q. What if we wanted to flip along the x -axis?

Amplitude-Reversal



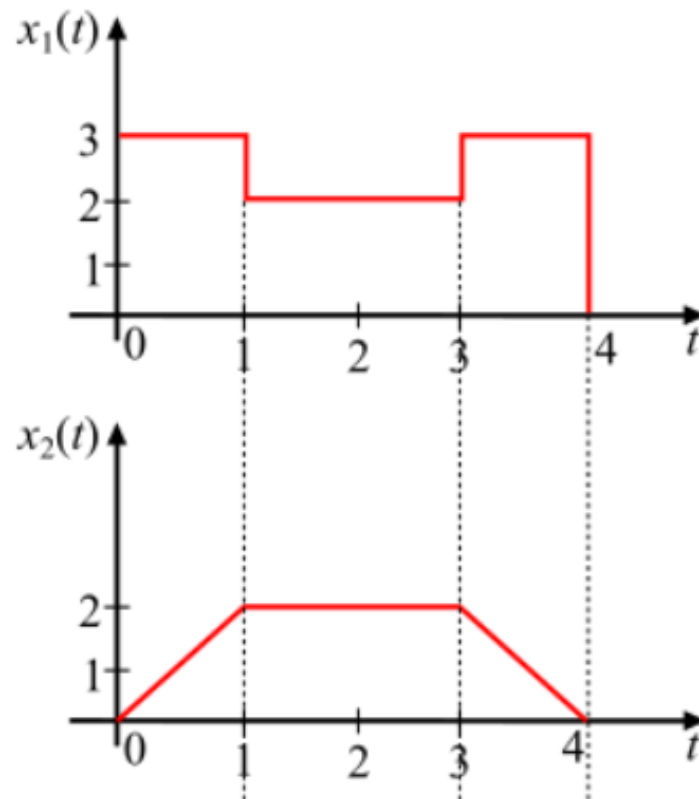
Q. What would the following do
 $x(t) + 5$, $x(t) - 5$, $5x(t)$?

Visualizing Time and Amplitude Reversal

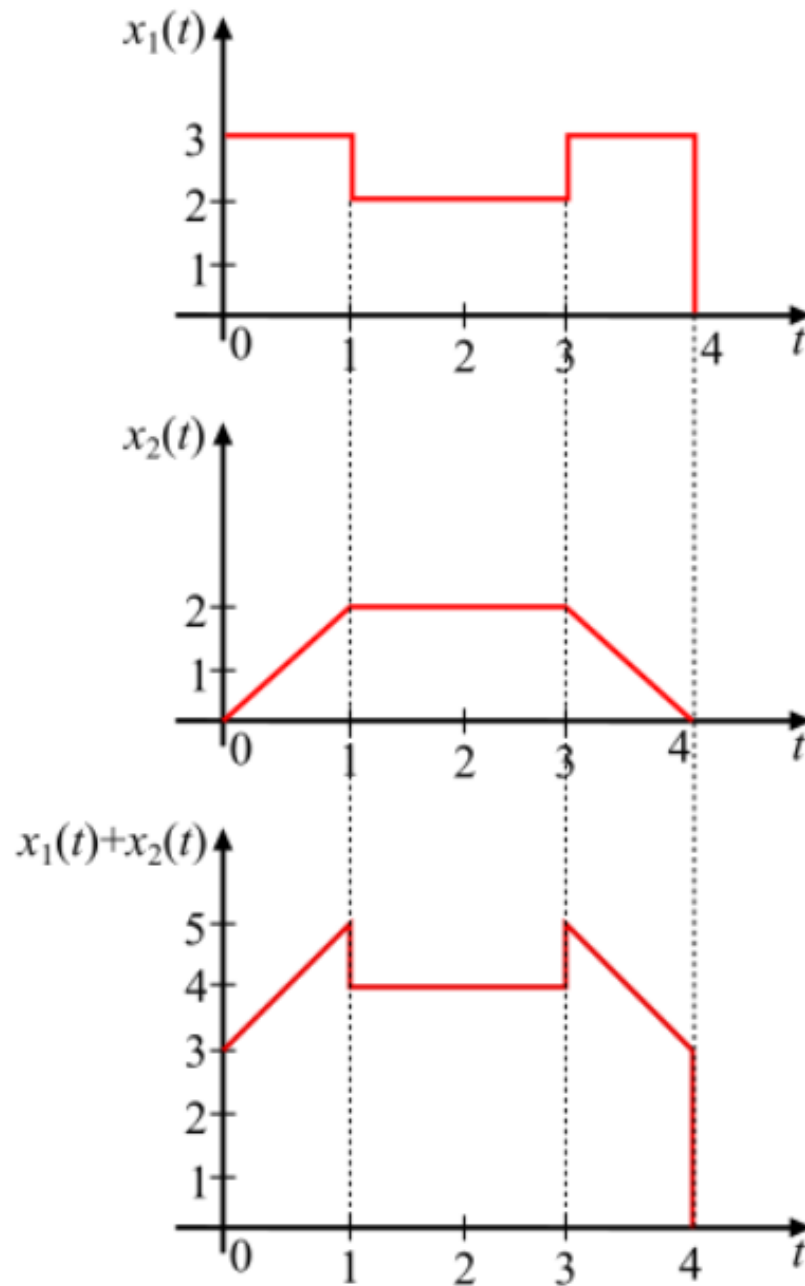


*Systems may take multiple
inputs and operate upon them.
Let's see a few simple
examples...*

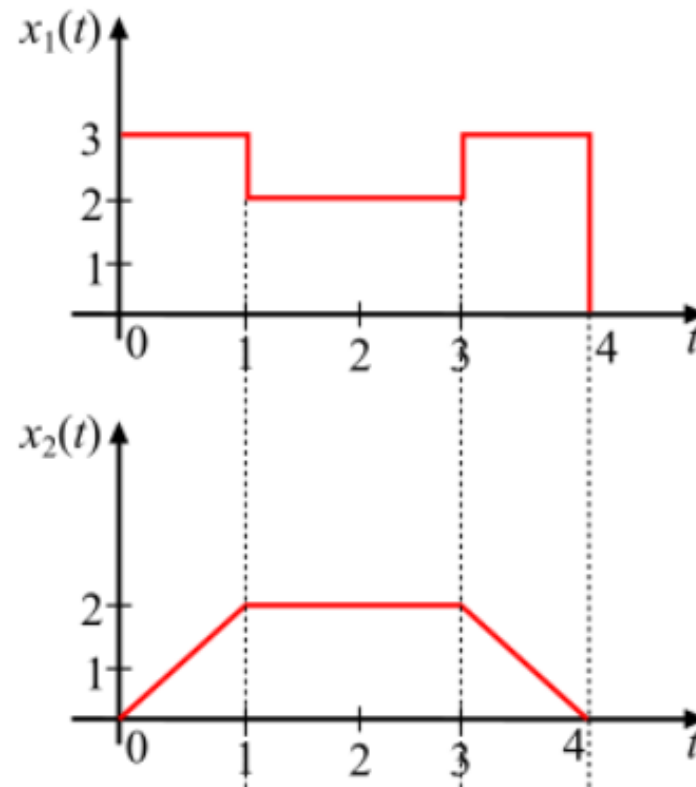
Q. Can you plot
 $x_1(t) + x_2(t)$?



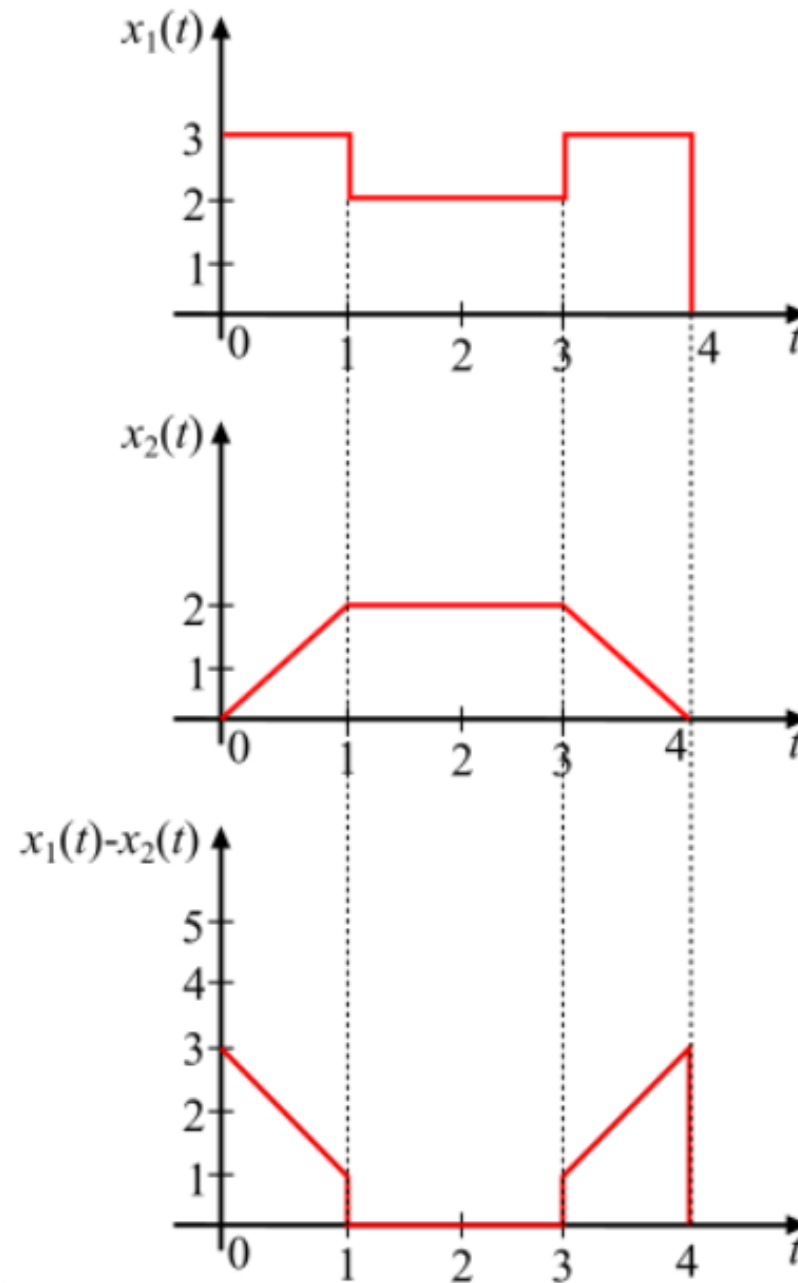
Q. Can you plot
 $x_1(t) + x_2(t)$?



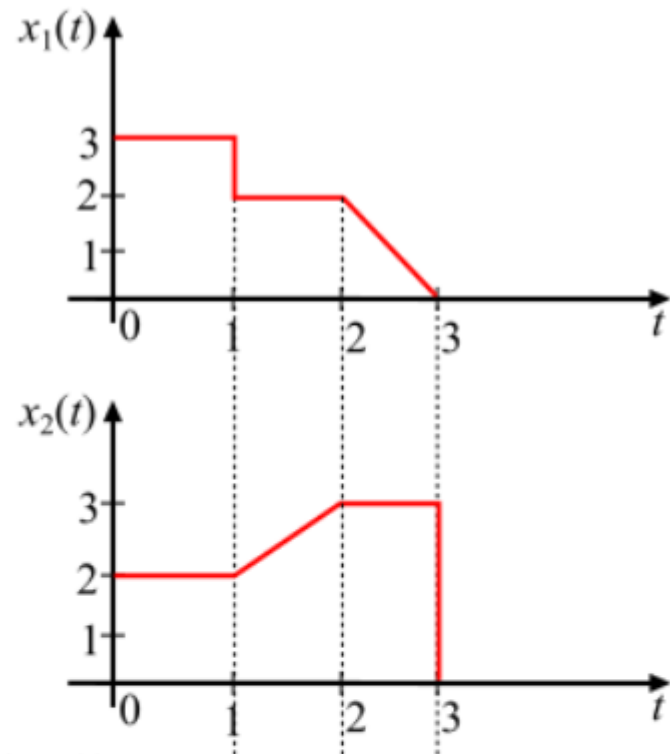
Q. Can you plot
 $x_1(t) - x_2(t)$?



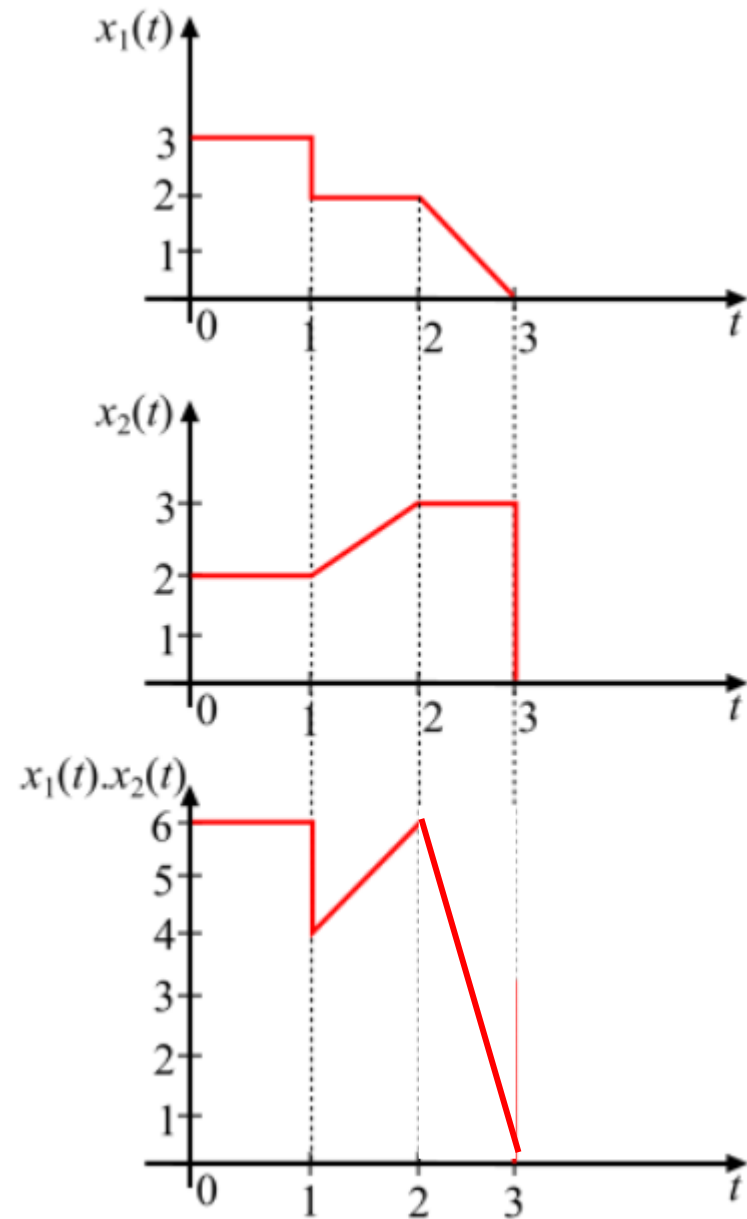
Q. Can you plot
 $x_1(t) - x_2(t)$?



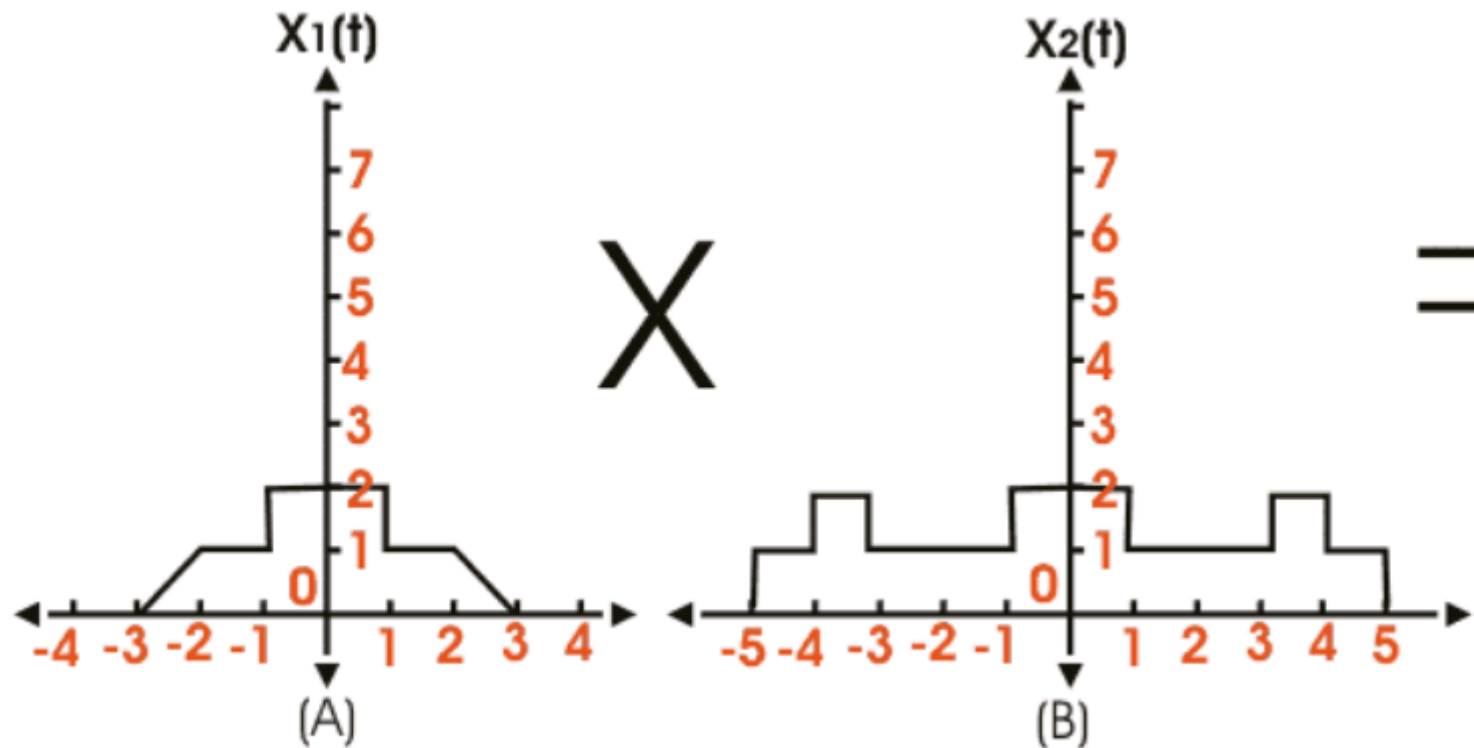
Q. Can you plot
 $x_1(t) \cdot x_2(t)$?



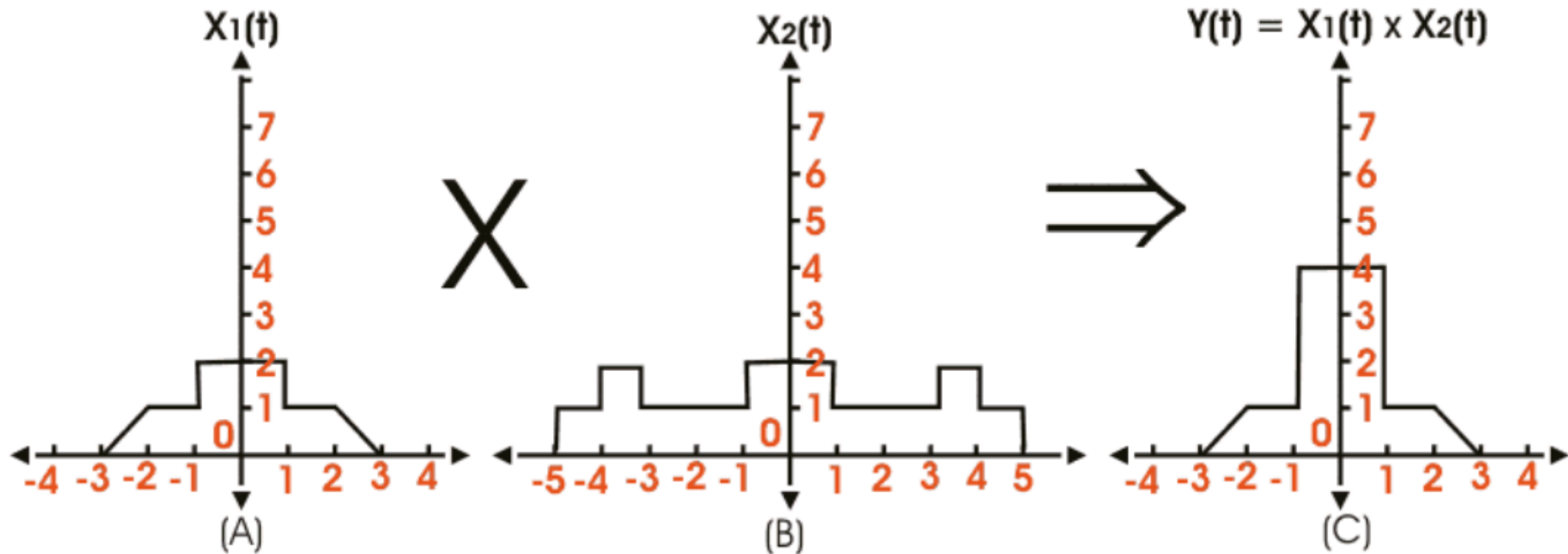
Q. Can you plot
 $x_1(t) \cdot x_2(t)$?



Q. Can you plot $x_1(t) \cdot x_2(t)$?



Q. Can you plot $x_1(t) \cdot x_2(t)$?



Systems often perform a multitude of operations on the signals entering them (e.g., combining the operations we saw above, and more...)

Q. Can you describe the operations of the following two systems mathematically?

System 1

I will delay the signal by 6 units and then compress by 2.

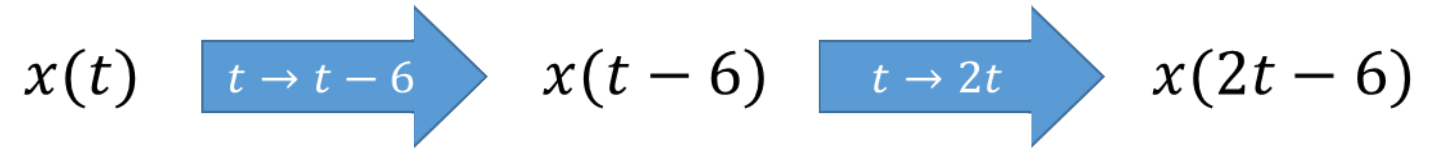
System 2

I will compress the signal by 2 and then delay by 3 units.

Q. Can you describe the operations of the following two systems mathematically?

System 1

I will delay the signal by 6 units and then compress by 2.



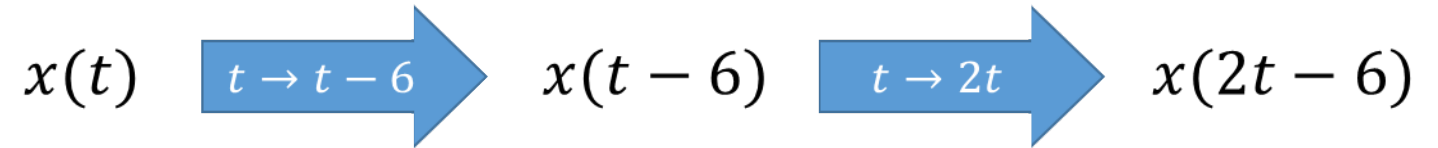
System 2

I will compress the signal by 2 and then delay by 3 units.

Q. Can you describe the operations of the following two systems mathematically?

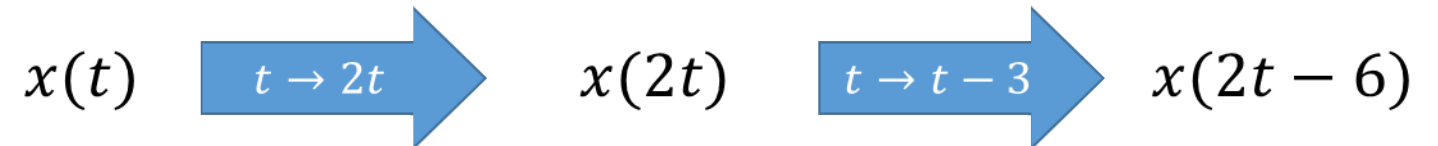
System 1

I will delay the signal by 6 units and then compress by 2.



System 2

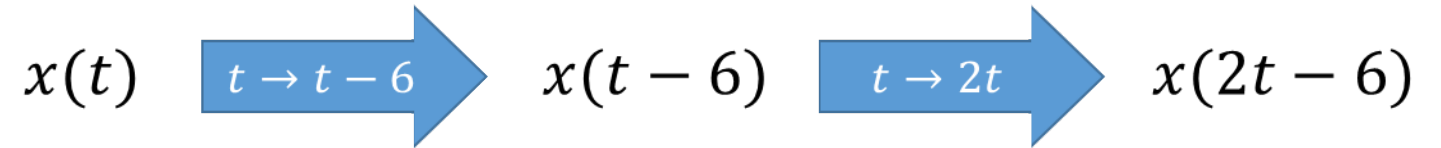
I will compress the signal by 2 and then delay by 3 units.



Q. Can you describe the operations of the following two systems mathematically?

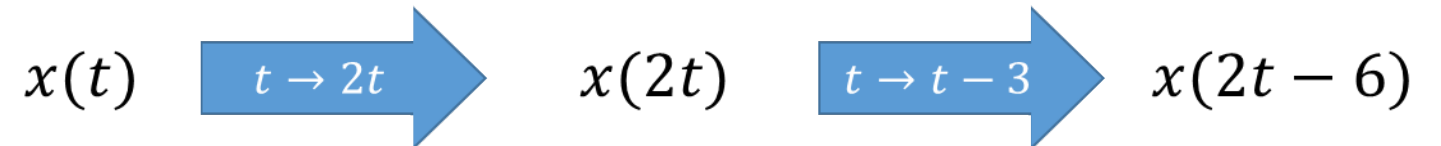
System 1

I will delay the signal by 6 units and then compress by 2.



System 2

I will compress the signal by 2 and then delay by 3 units.



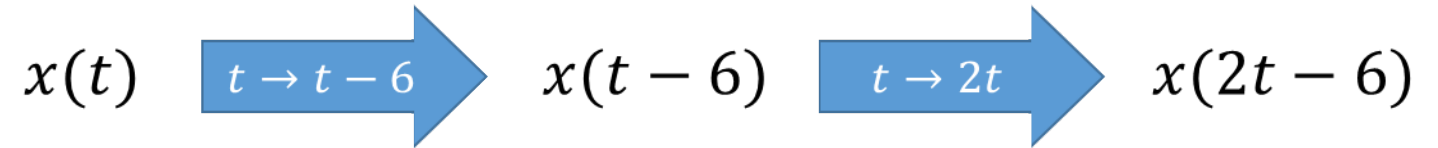
- Note that the two systems perform the same net operation.

Lesson

Q. Can you describe the operations of the following two systems mathematically?

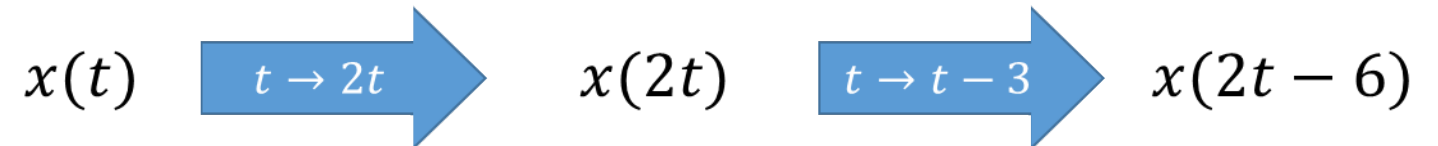
System 1

I will delay the signal by 6 units and then compress by 2.



System 2

I will compress the signal by 2 and then delay by 3 units.



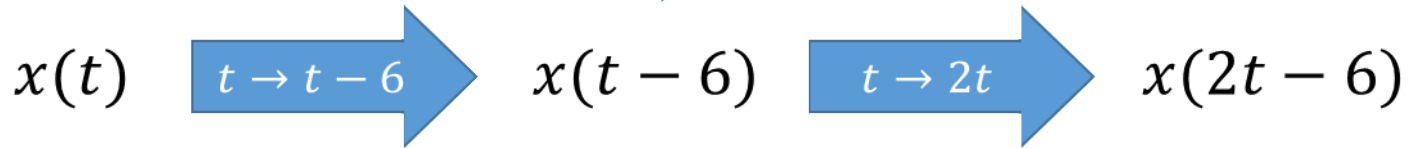
Lesson

- Note that the two systems perform the same net operation.
- However, the second is a bit confusing due to scaling done first.

Q. Can you describe the operations of the following two systems mathematically?

System 1

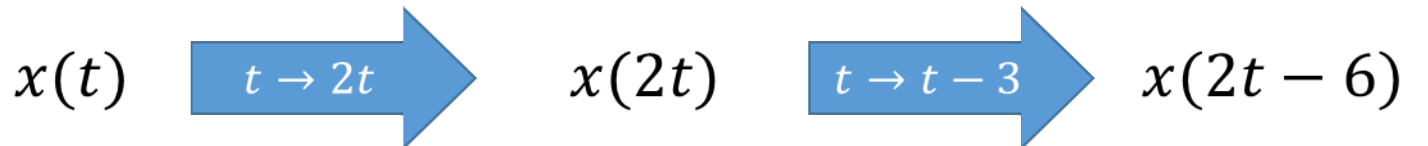
I will delay the signal by 6 units and then compress by 2.



Warning: mistakenly replacing $t - 6 \rightarrow 2(t - 6)$ here would lead to compression *and* additional translation!

System 2

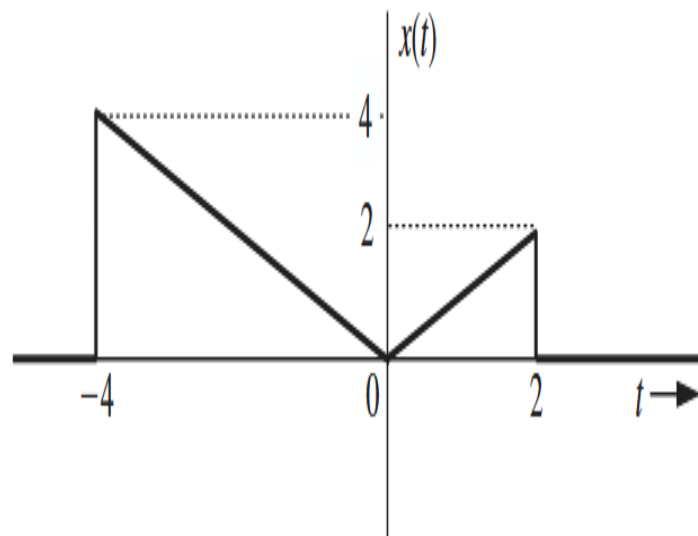
I will compress the signal by 2 and then delay by 3 units.



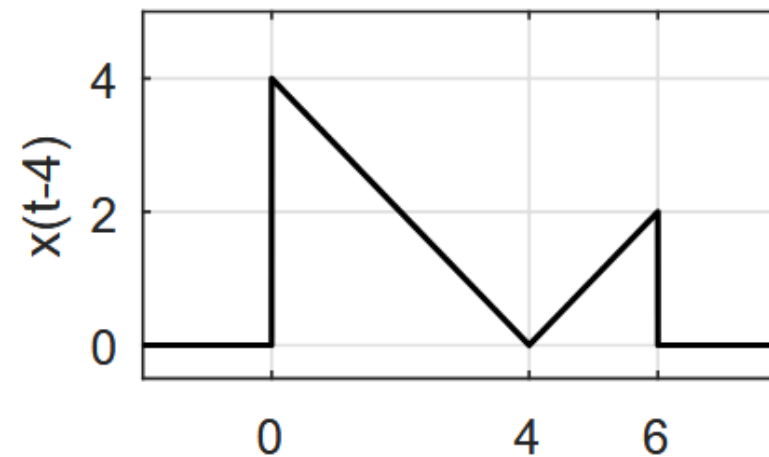
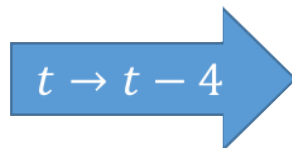
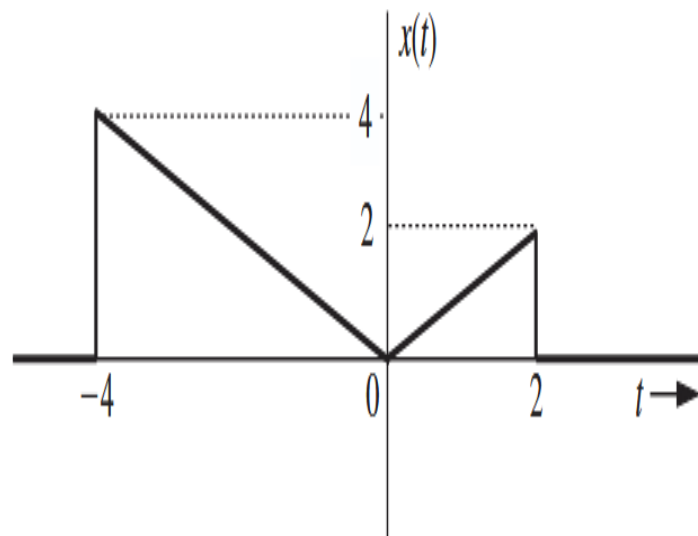
Lesson

- Note that the two systems perform the same net operation.
- However, the second is a bit confusing due to scaling done first.
- **It is, therefore, advisable to perform shifting first and then carry out any scaling or time-reversal!**

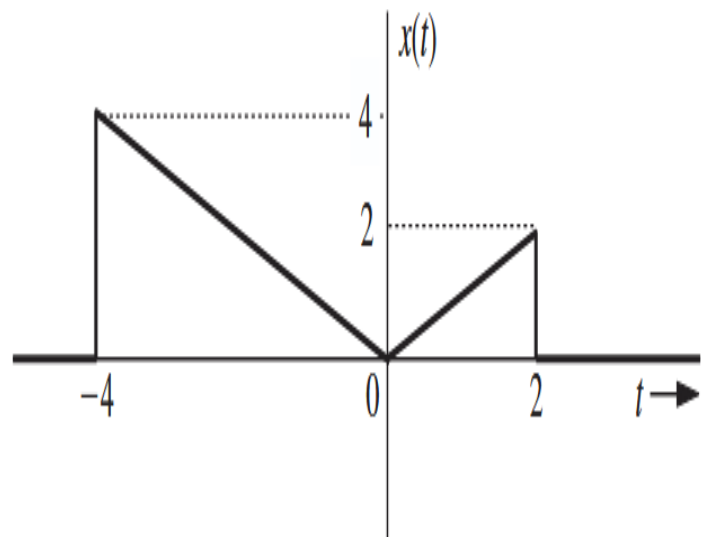
Given $x(t)$, plot $x(2t - 4)$



Given $x(t)$, plot $x(2t - 4)$



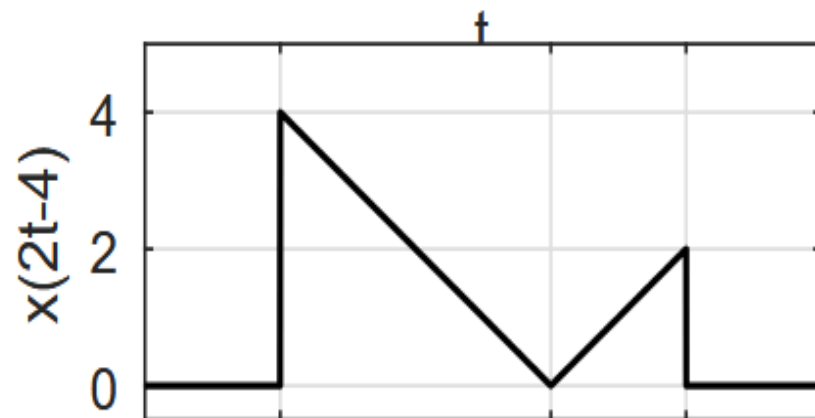
Given $x(t)$, plot $x(2t - 4)$



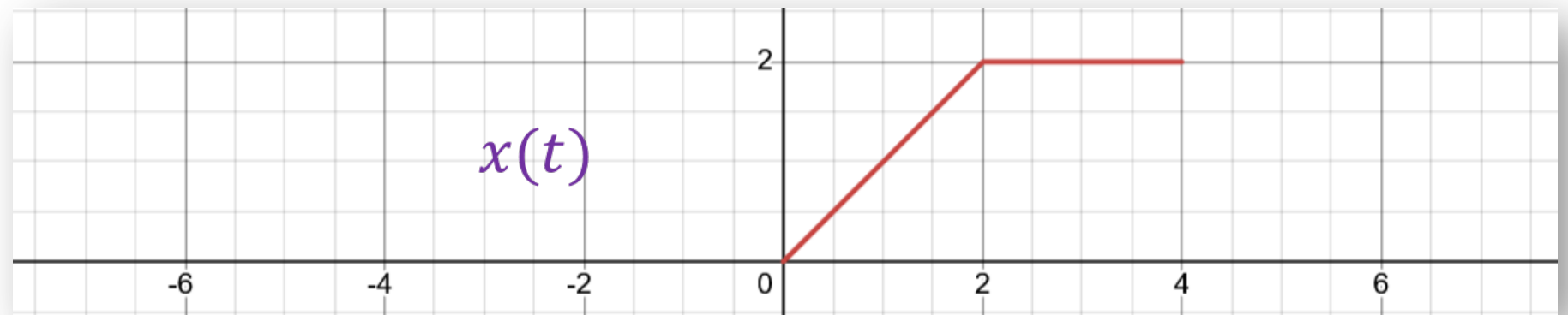
$t \rightarrow t - 4$



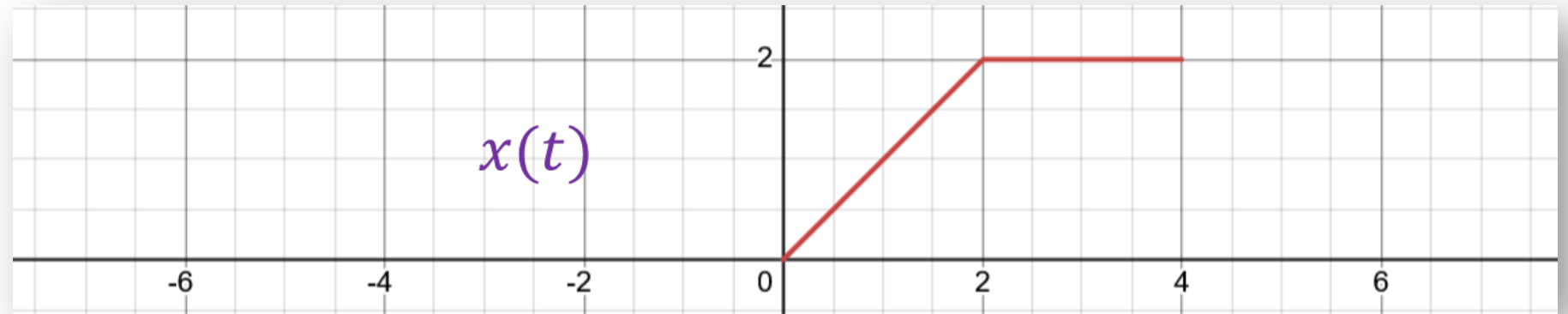
$t \rightarrow 2t$



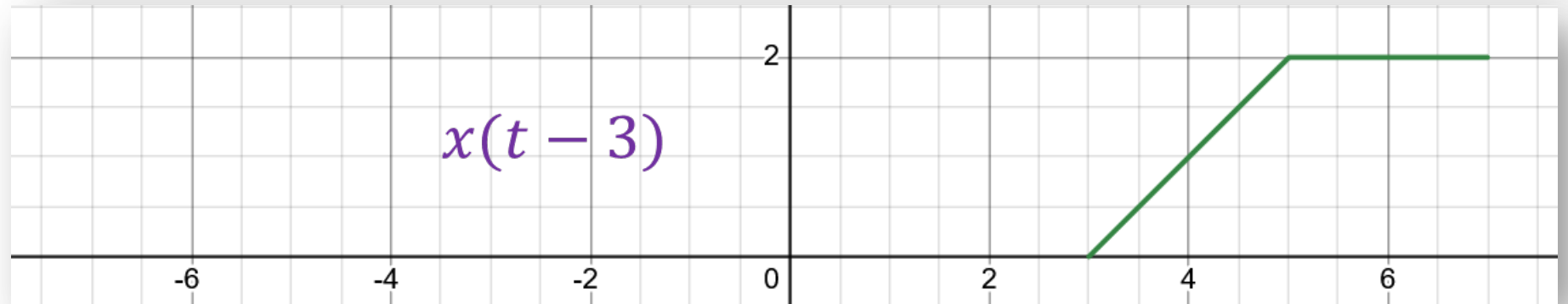
Given $x(t)$,
plot $x(-t - 3)$



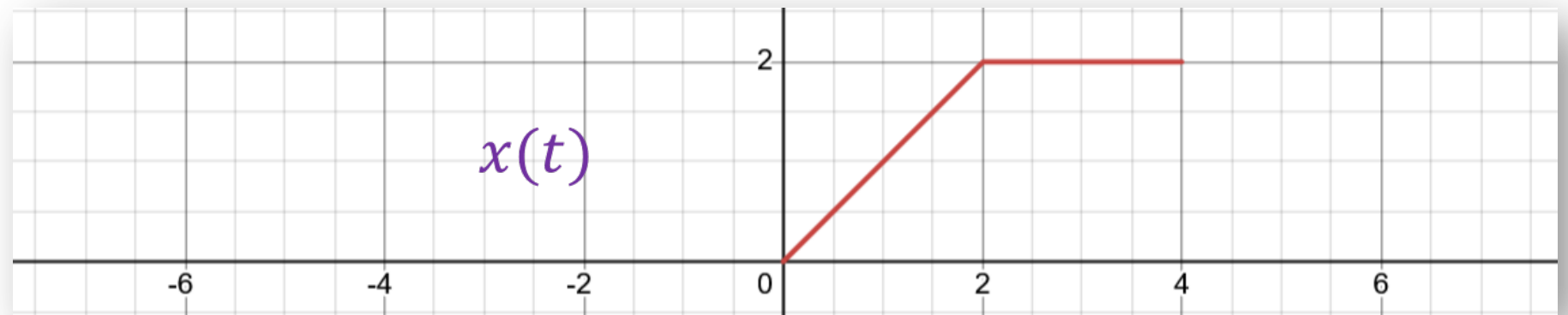
Given $x(t)$,
plot $x(-t - 3)$



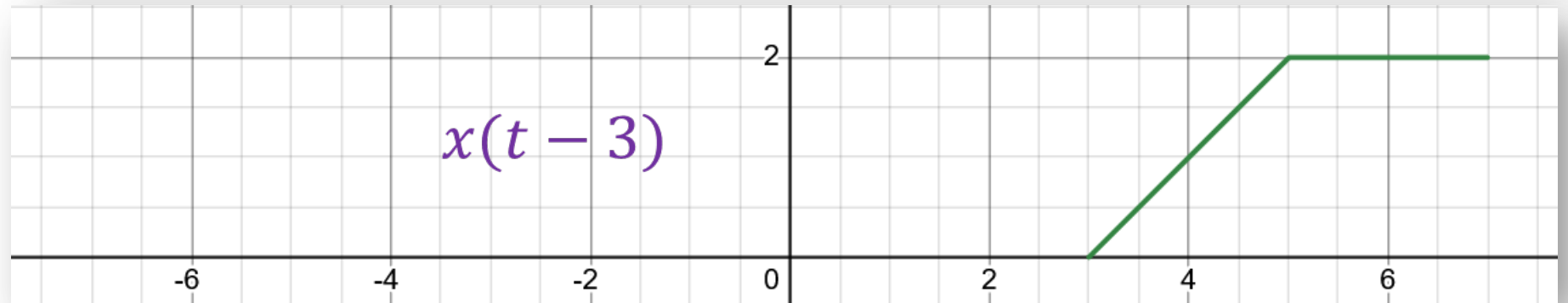
$t \rightarrow t - 3$



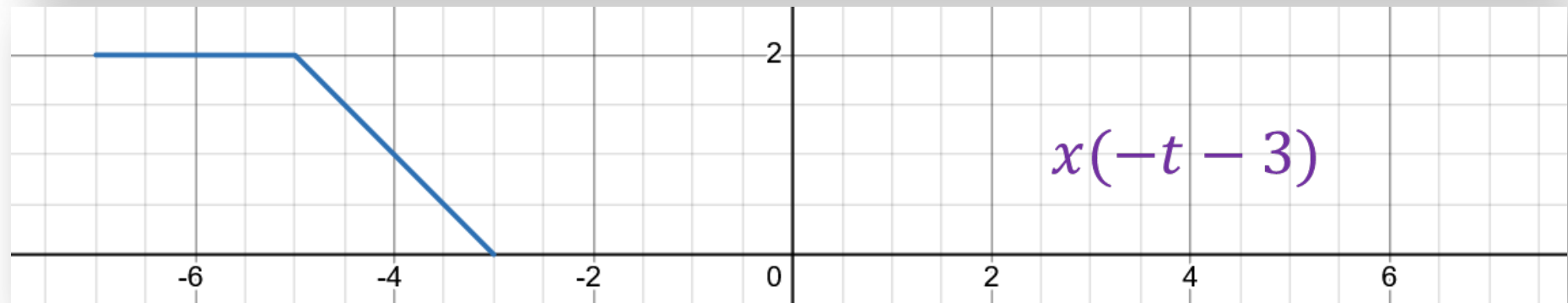
Given $x(t)$,
plot $x(-t - 3)$



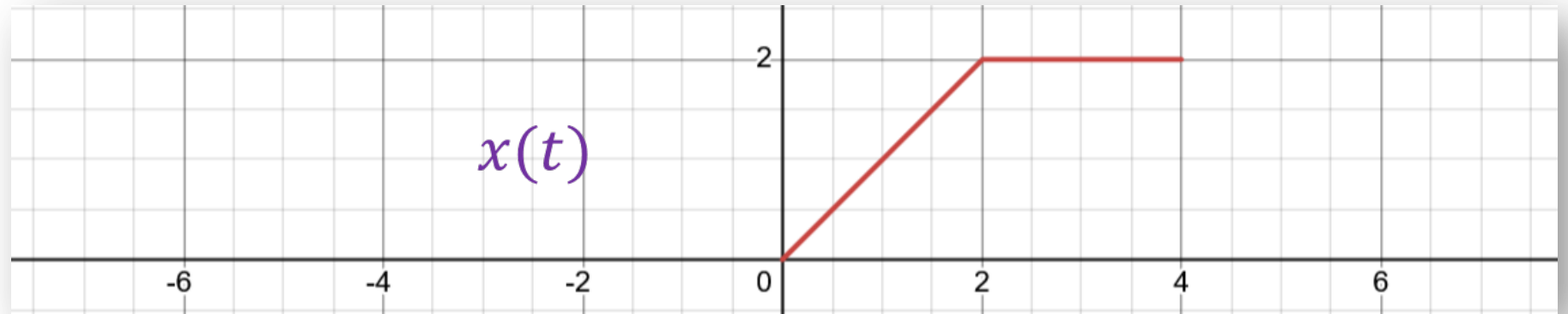
$t \rightarrow t - 3$



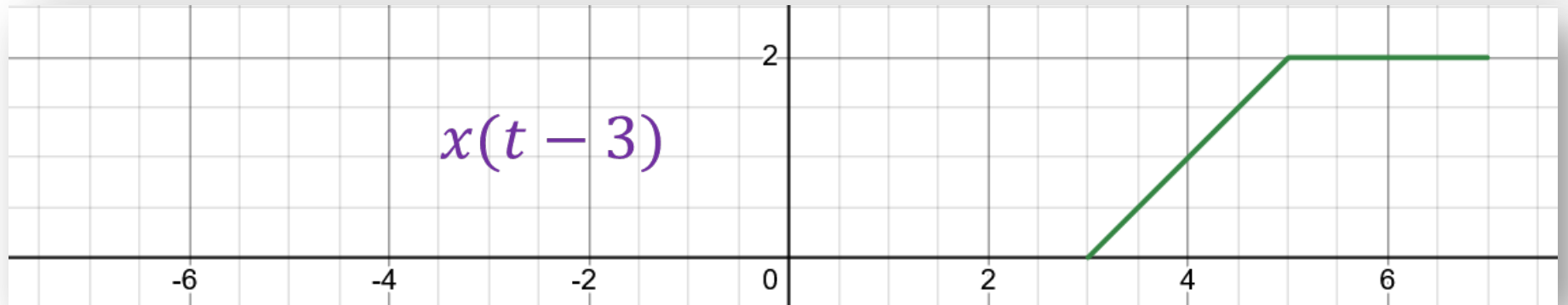
$t \rightarrow -t$



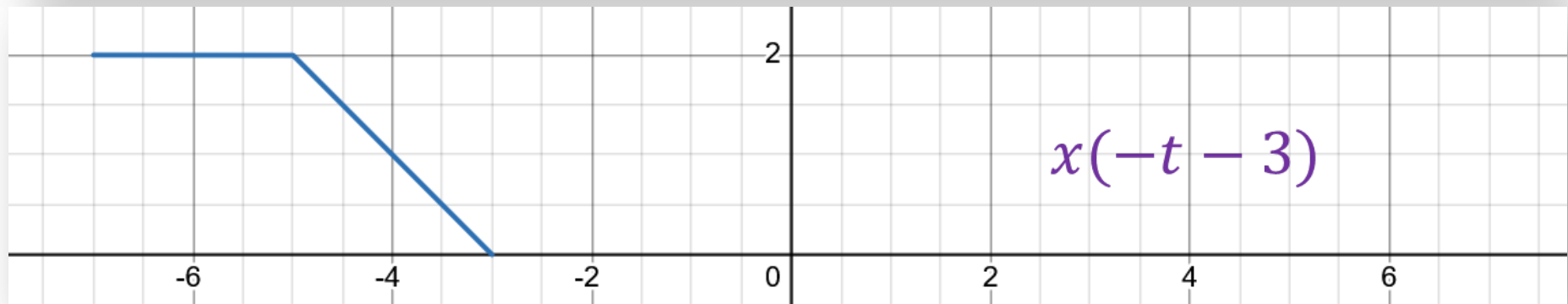
Given $x(t)$,
plot $x(-t - 3)$



$t \rightarrow t - 3$



$t \rightarrow -t$



- Interestingly, while -3 in $x(t-3)$ is a delay, -3 in $x(-t-3)$ is an advance.
- To avoid any confusion, we carry out shifting first.

Questions?? Thoughts??

